

Task 4: Schema Control and Administration (A)

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1372-Information Technology Business Analysis, Conestoga College – Doon Campus

PROG8650: W23-SEC 2 Database Management Systems

Yasser Elmankabady

Assignment Due Date: 07/04/2023

EMAIL 1:

--Using the existing Employee Test Database(Group Project 2 Database) having dbo schema.

- EMPLOYEE (EmployeeID, RoleID, ManagerID, DepartmentID, EmpAddressID, FName, LName, Salary, HireDate, Probation, PersonalEmail, PersonalPhoneNumber)
- ROLES (RoleID, PositionTitle, PositionDescription, ProbationLength, MinSalary, MaxSalary)
- DEPENDENTS (DependentID, EmployeeID, Relationships, FName, LName, DateOfBirth)
- EMPLOYEEPERSONAL (PersonalID, EmployeeID, DateOfBirth, SocialSecurityNumber, AllergyName, AllergyDesc, EpiPen, DisabilityName, DisabilityDesc, Gender)
- DEPARTMENTS (DepartmentID, FacilityID, DepartmentName, DepartmentDesc)
- FACILITYADDRESS (FacilityID, CountryID, FacilityName)
- EMPLOYEECONTACT (EmpContactID, EmployeeID, FacilityID, WorkExtension, OfficeNumber)
- EMERGENCYCONTACTS (ContactID, EmployeeID, FName, LName, ContactNumber, ContactAltNumber)
- ADDRESS (AddressID, Facility, StreetNumber, StreetName, StreetType, PostalCode, City, Province)
- COUNTRIES (CountryID, RegionID, CountryName)
- EMPLOYEEADDRESS (EmpAddressID, CountryID, HomeType, UnitNumber)
- REGIONS (RegionID, RegionName)

USE EmployeeTest

GO

-- Creating new schema EmployeeRole and changing the dbo.schema to EmployeeRole

CREATE SCHEMA EmployeeRole AUTHORIZATION dbo;

ALTER SCHEMA EmployeeRole

TRANSFER dbo.EMPLOYEE;

ALTER SCHEMA EmployeeRole

TRANSFER dbo.ROLES;

-- Creating new schema EmployeePersonalData and changing the dbo.schema to EmployeePersonalData

CREATE SCHEMA EmployeePersonalData AUTHORIZATION dbo;

ALTER SCHEMA EmployeePersonalData

TRANSFER dbo.EMPLOYEEPERSONAL;

ALTER SCHEMA EmployeePersonalData

TRANSFER dbo.DEPENDENTS;

-- Creating new schema Facility and changing the dbo.schema to Facility

CREATE SCHEMA Facility AUTHORIZATION dbo;

ALTER SCHEMA Facility

TRANSFER dbo.DEPARTMENTS;

-- Creating new schema Contact and changing the dbo.schema to Contact

CREATE SCHEMA Contact AUTHORIZATION dbo;

ALTER SCHEMA Contact

TRANSFER dbo.EMPLOYEECONTACT;

ALTER SCHEMA Contact

TRANSFER dbo.EMERGENCYCONTACTS;

-- Creating new schema FullAddress and changing the dbo.schema to FullAddress

CREATE SCHEMA FullAddress AUTHORIZATION dbo;

```
ALTER SCHEMA FullAddress  
TRANSFER dbo.EMPLOYEEADDRESS;
```

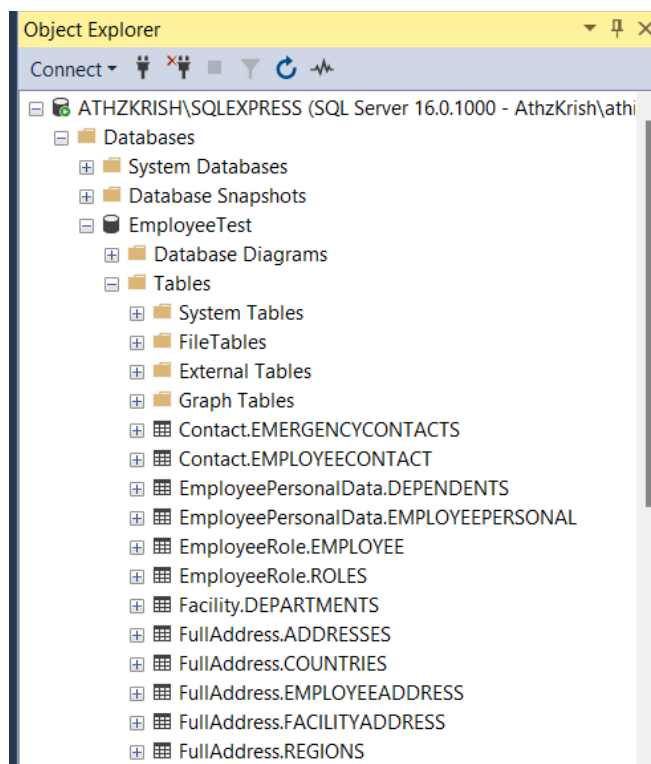
```
ALTER SCHEMA FullAddress  
TRANSFER dbo.FACILITYADDRESS;
```

```
ALTER SCHEMA FullAddress  
TRANSFER dbo.ADDRESSES;
```

```
ALTER SCHEMA FullAddress  
TRANSFER dbo.COUNTRIES;
```

```
ALTER SCHEMA FullAddress  
TRANSFER dbo.REGIONS;
```

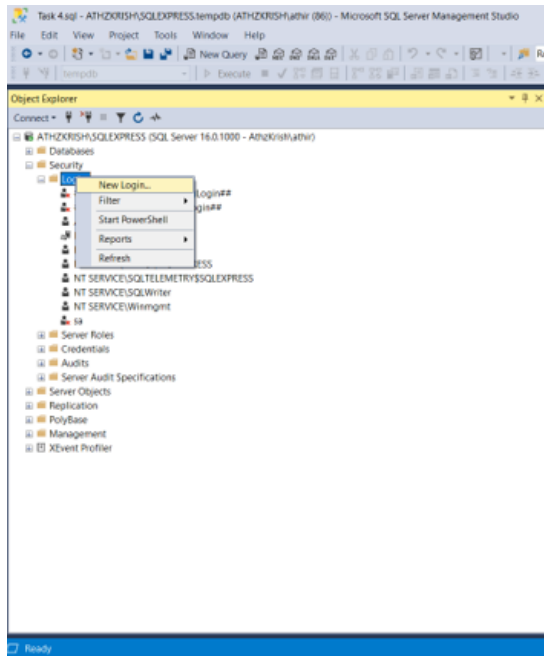
The screenshot of schemas in EmployeeTest DB after executing above statements:



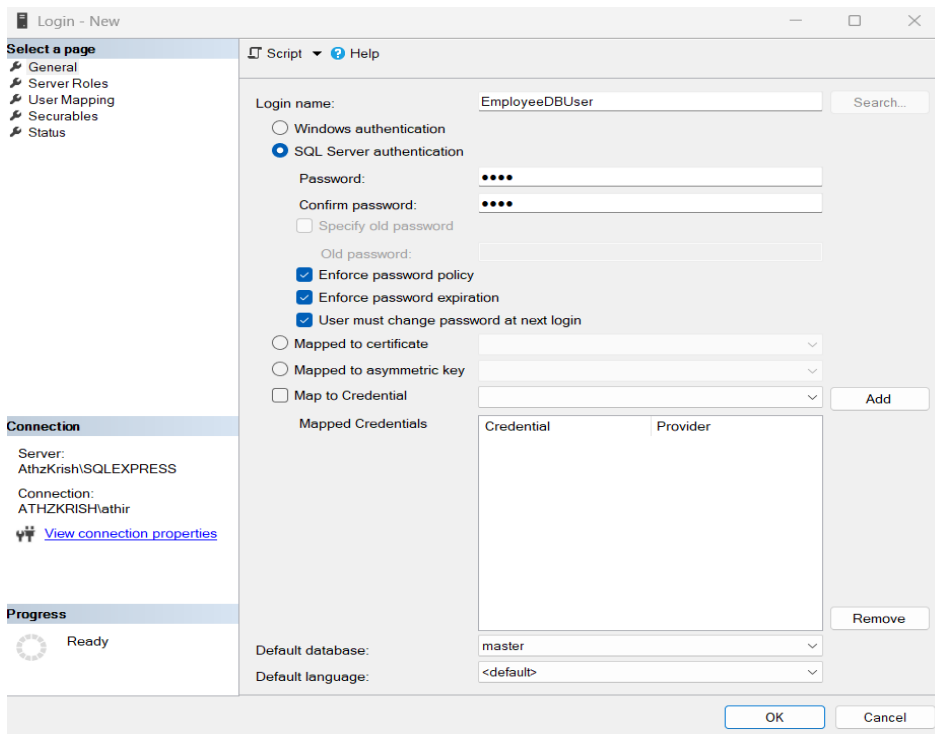
SETTING UP ADMINISTRATOR USER ACCOUNT BY CREATING SERVER LOGIN AND DATABASE USER:

I. Create Login to the My SQL Server Instance -Server Level Login

1. Log in to My SQL Server.
2. Go to Security of My SQL Server Instance-> Login and click on New Login option.

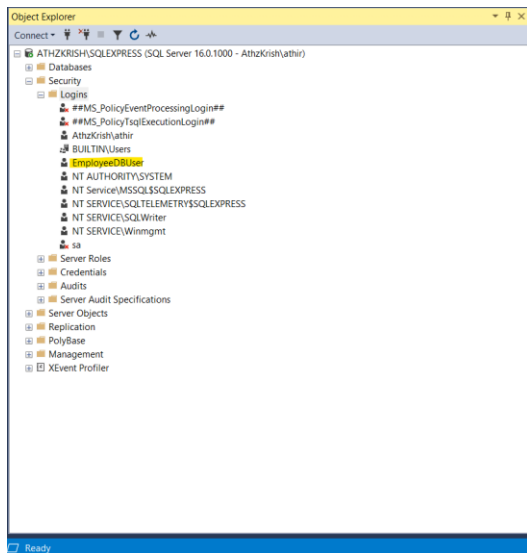


New Login creation window opens as below:



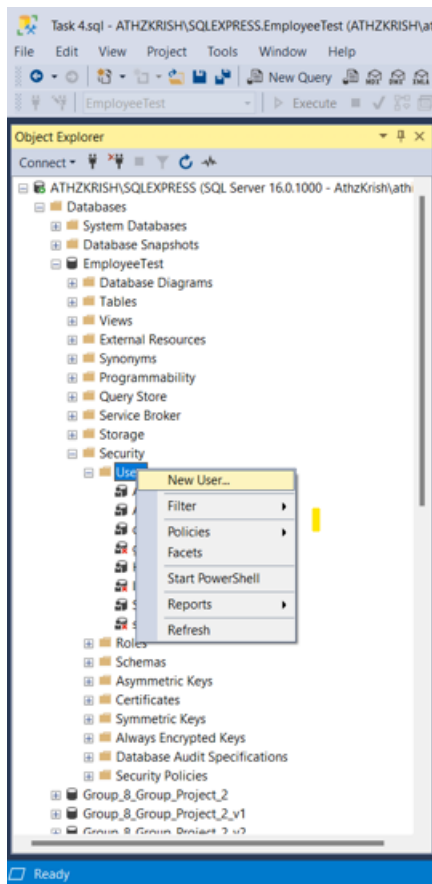
3. Enter the Login Name.
4. Select SQL Server authentication option(For creating logins with password).
5. Enter new password for the login in Password and Confirm Password textboxes.
6. Click OK .

EmployeeDBUser Login got created as below:



II. Create the Database Level user logins for the Employee Database(Database Name: EmployeeTest)

1. Go to the above database name.
2. Go to Security->User and click on New User option.



New User creation window opens:

3. Enter the User Name.
Here, I'm creating the Employee Administrator User Account. Hence, given the user name as 'Administrator'.
Click on the ellipsis(...)next to the Login name textbox to select the server login for this new Administrator user

Database User - New

Select a page

- General
- Owned Schemas
- Membership
- Securables
- Extended Properties

Script Help

User type:
SQL user with login

User name:
Administrator

Login name:

Default schema:

Connection

Server:
AthzKrish\SQLEXPRESS

Connection:
ATHZKRISH\athir

[View connection properties](#)

Progress

Ready

OK Cancel

The below opens:

Select Login

Select these object types:

Logins

Object Types...

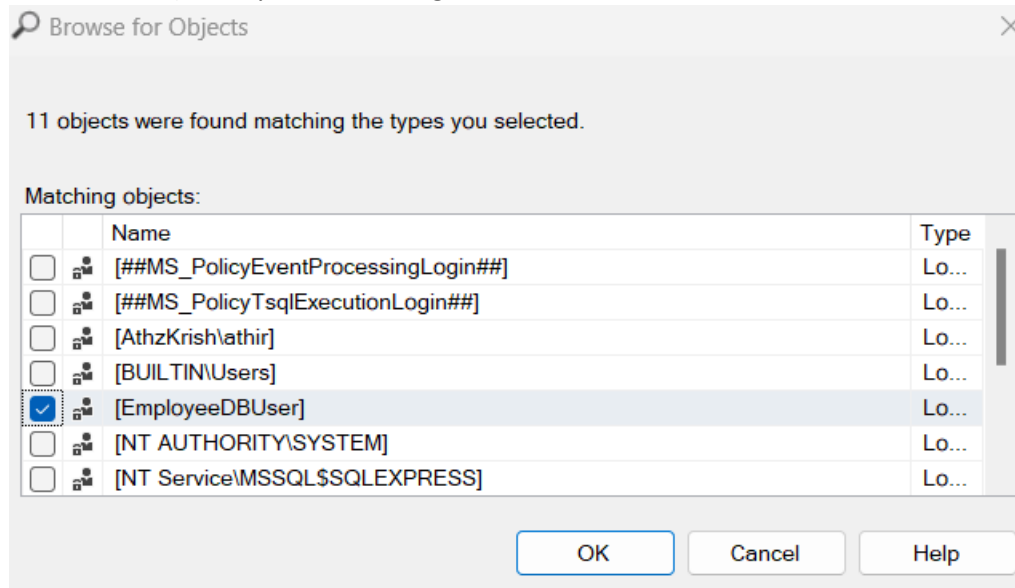
Enter the object names to select (examples):

Check Names

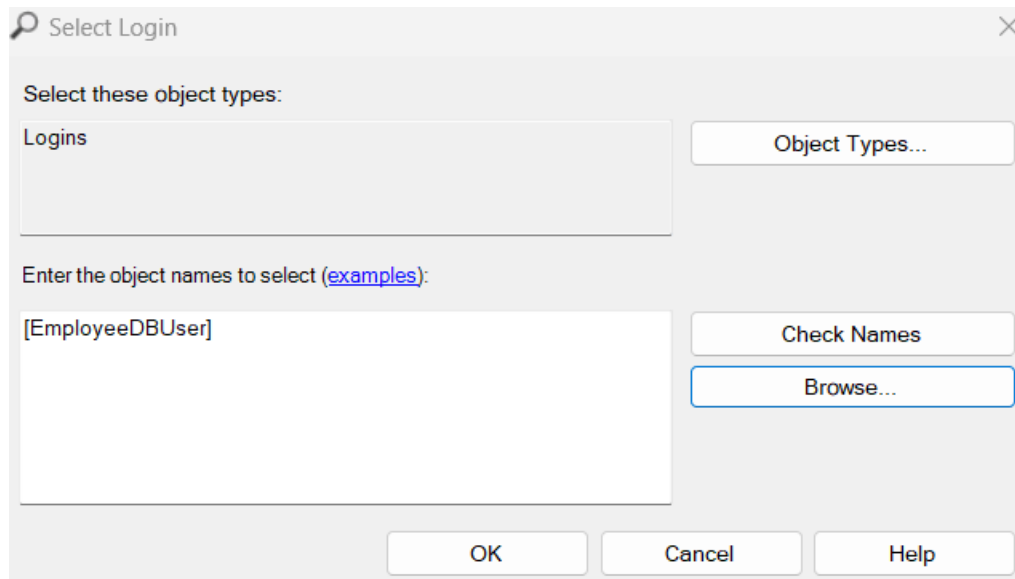
Browse...

OK Cancel Help

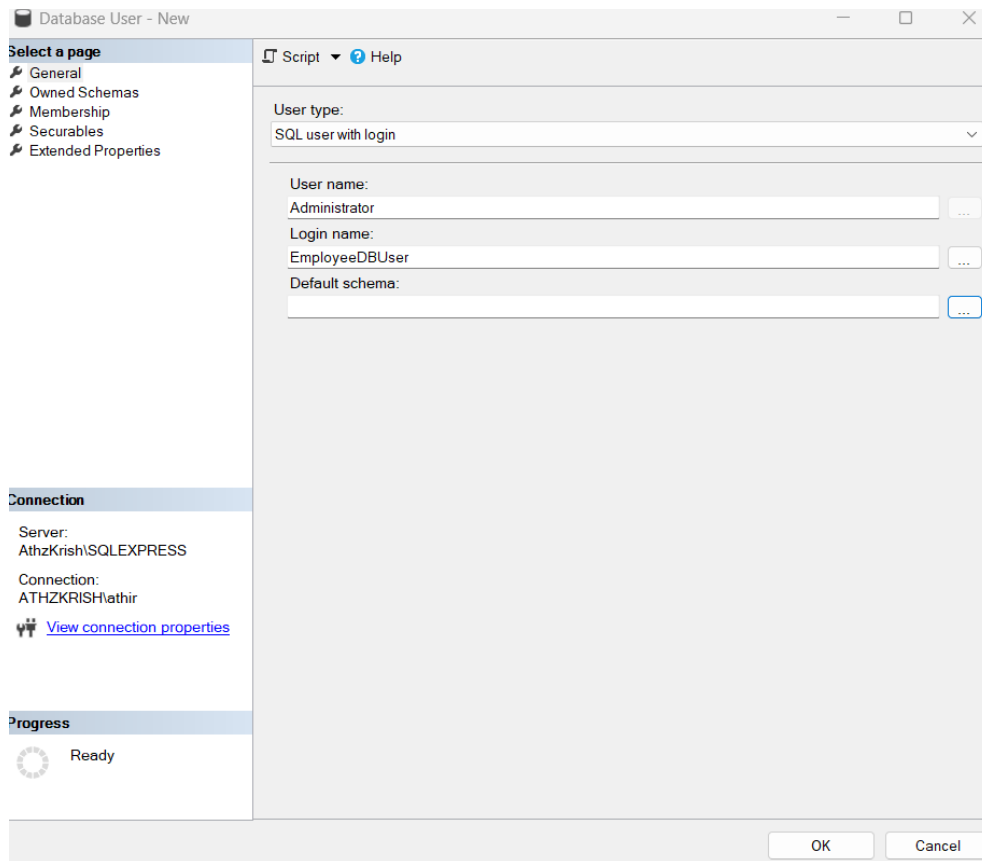
4. Click on Browse button to select the Server Login for this user. Select the EmployeeDBUser Object from the list and Click OK. (The My SQL server login created for the Administrator User Account of the Employee Database)



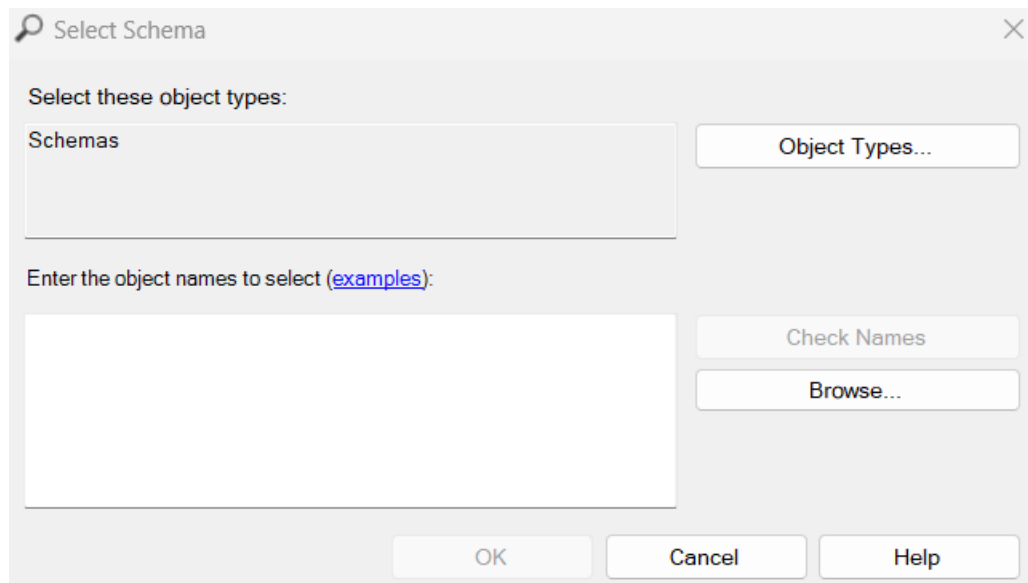
5. The selected object appears under object names. Click OK



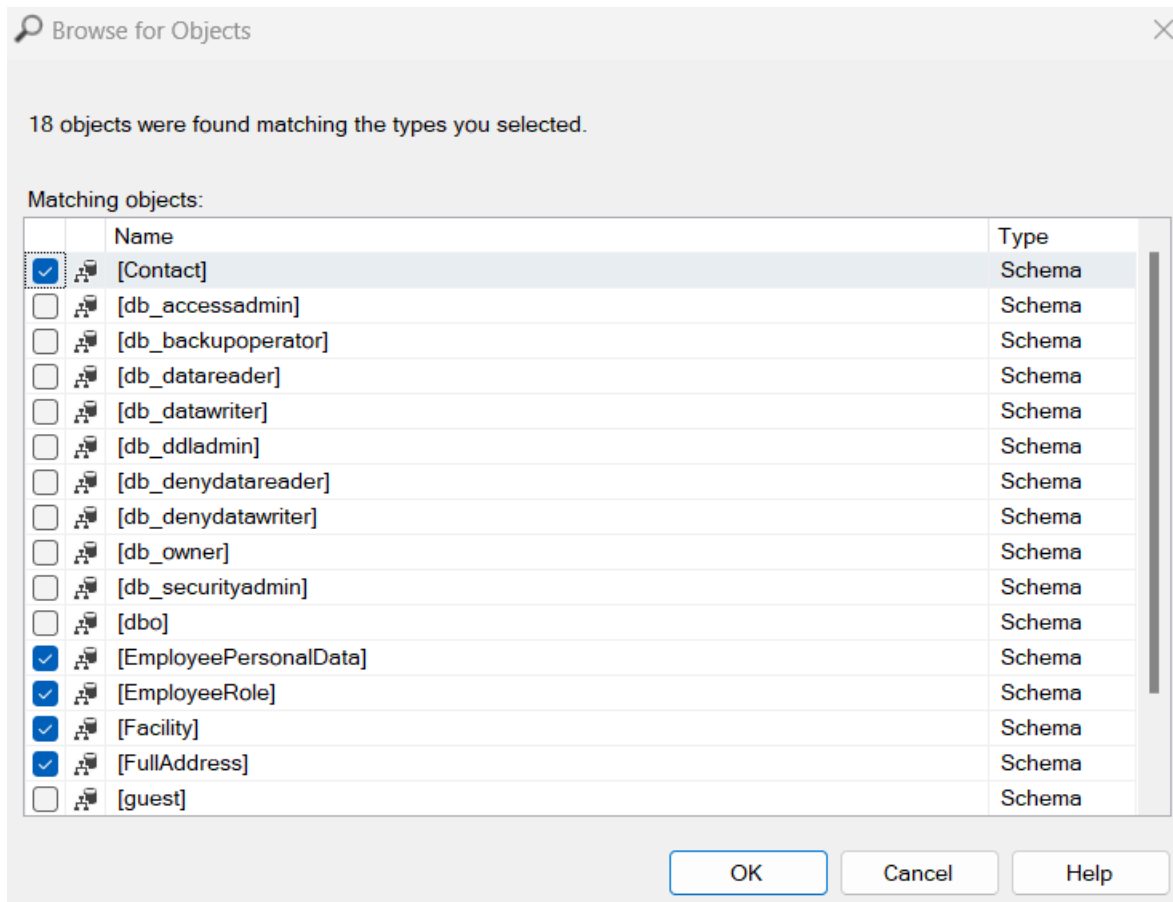
6. Now, select the schema which the user needs to access by clicking on the ellipsis(...) next to the Default schema textbox.



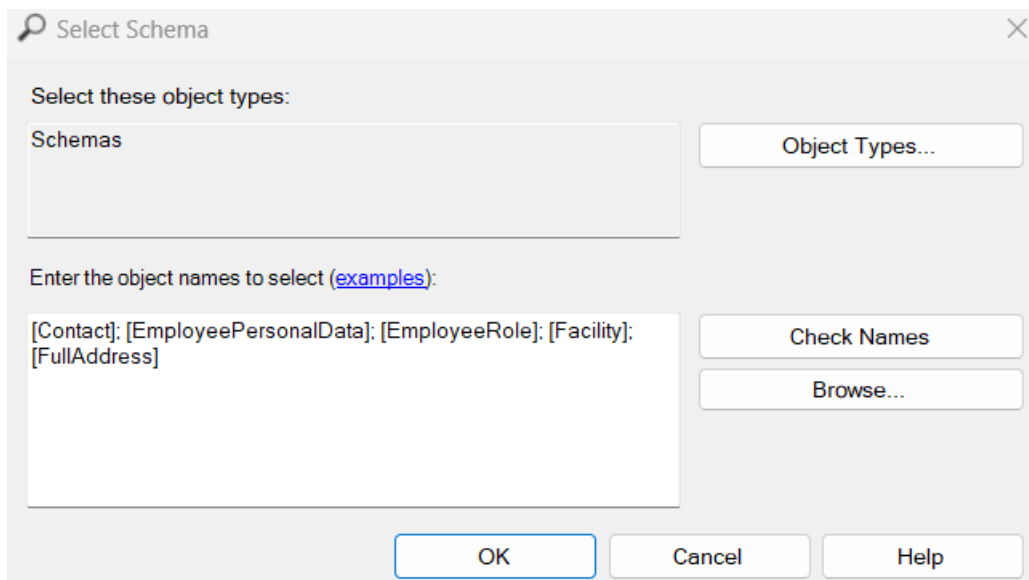
The below window appears:



- Click on the Browse button and select all 5 schemas: EmployeeRole, EmployeePersonalData, Facility, Contact, and FullAddress of Employee Database.



8. Selected schemas were added under the object names. Click OK.



Database User - New

Script Help

Select a page

- General
- Owned Schemas
- Membership
- Securables
- Extended Properties

Connection

Server:
AthzKrish\SQLEXPRESS

Connection:
ATHZKRISH\athir

[View connection properties](#)

Progress

Ready

User type:
SQL user with login

User name:
Administrator

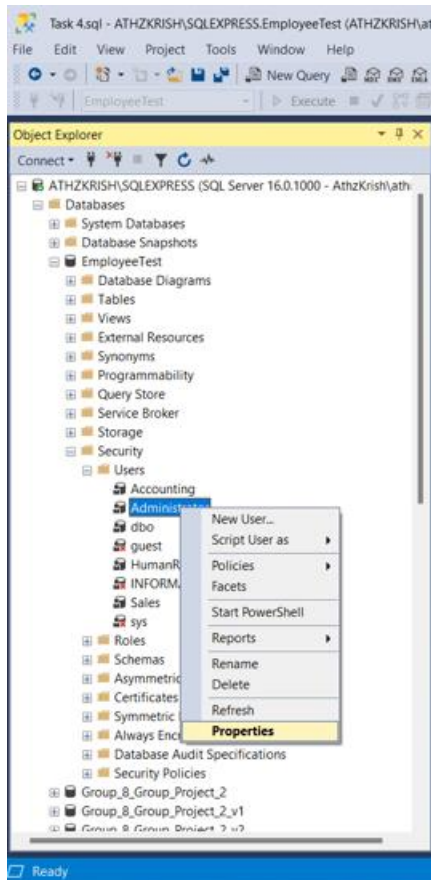
Login name:
EmployeeDBUser

Default schema:
Contact

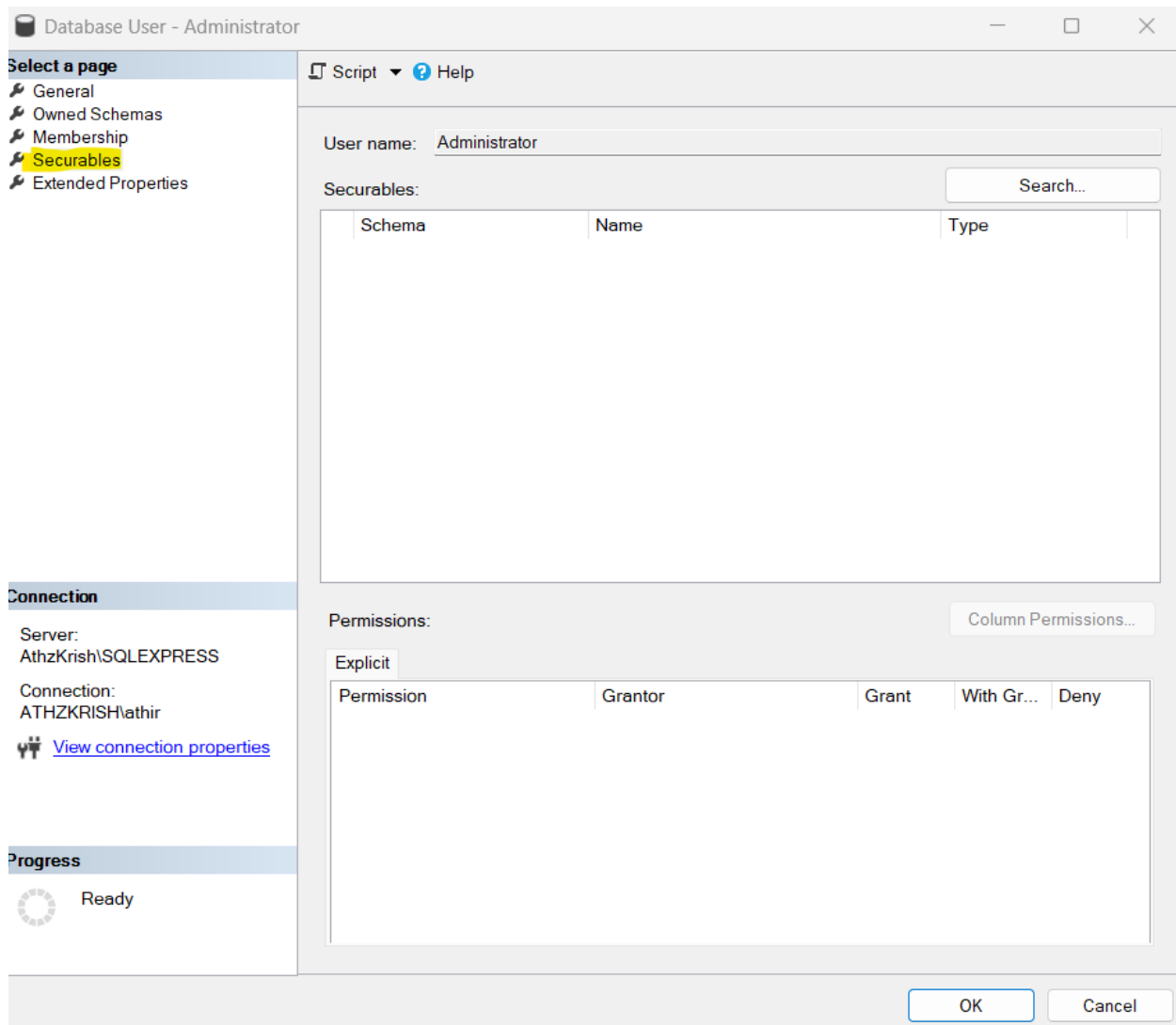
OK Cancel

III. Adding Permissions for Administrator User Account

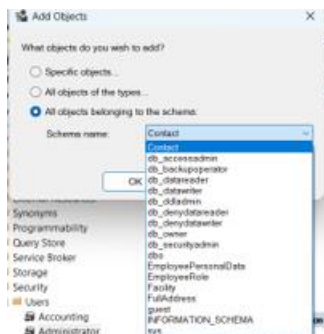
1. Go back to Database name ->Security->Users->Administrator and click on Properties.



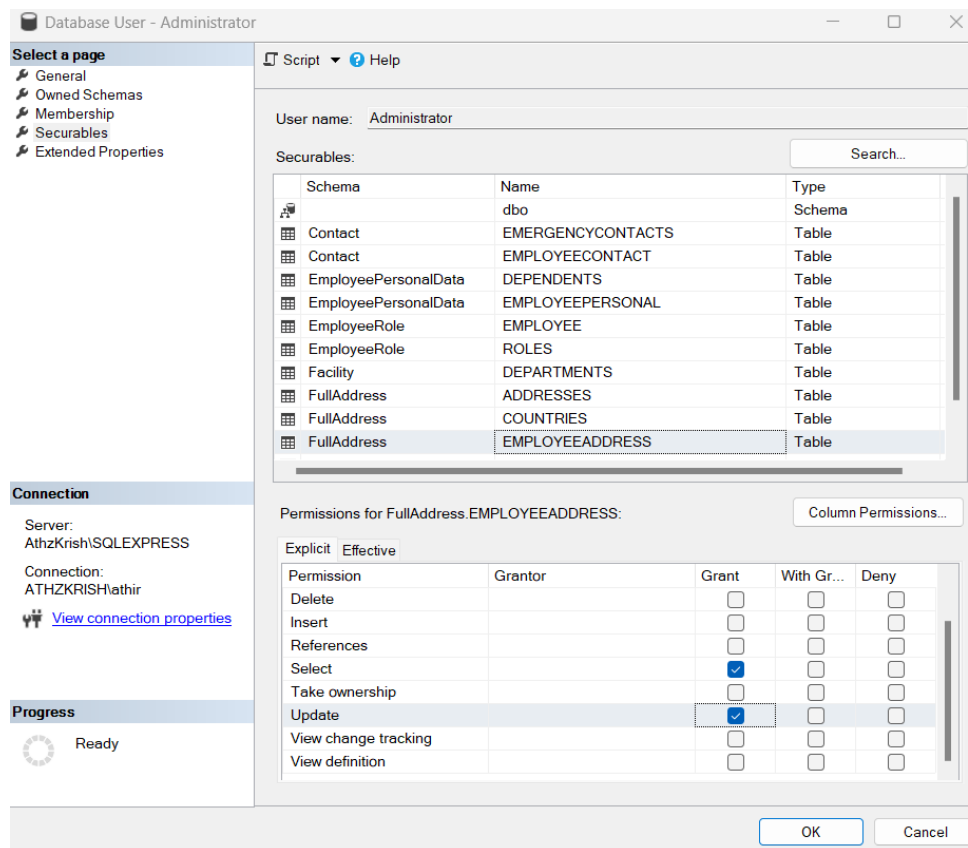
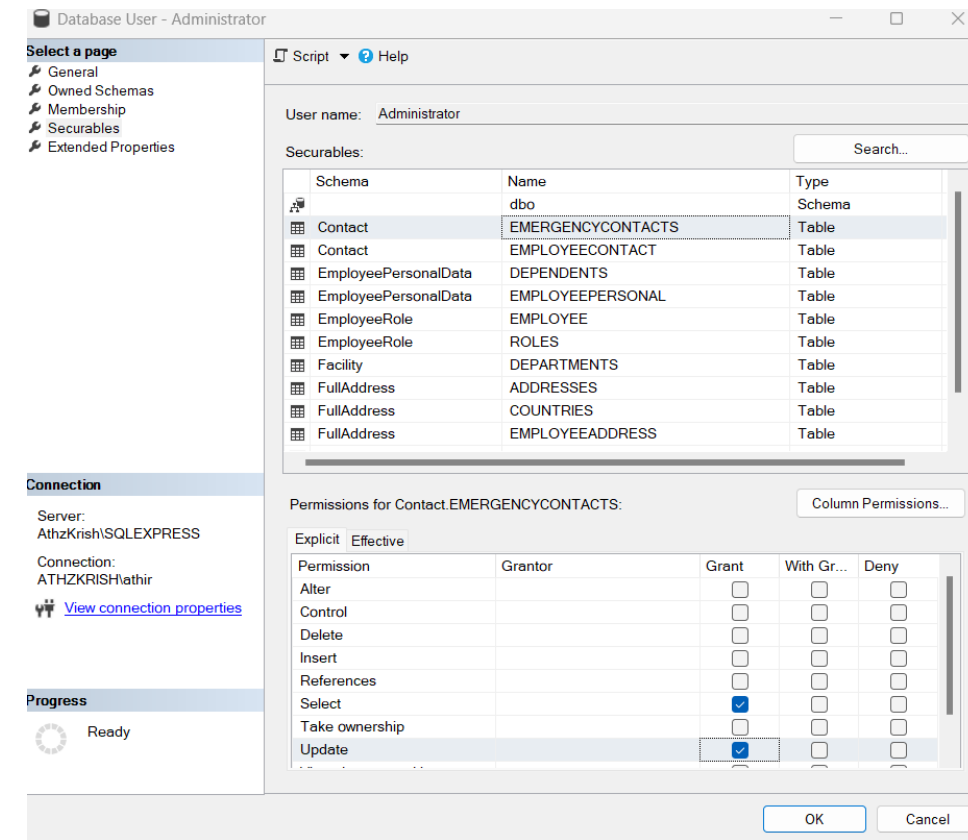
2. Select Securables from the left and click on the Search button:



- Click 'All objects belonging to the schema' for the Administrator user in order to access all the schemas under the Employee Database, select all 5 schemas: EmployeeRole, EmployeePersonaData, Facility, Contact, and FullAddress, and click OK.

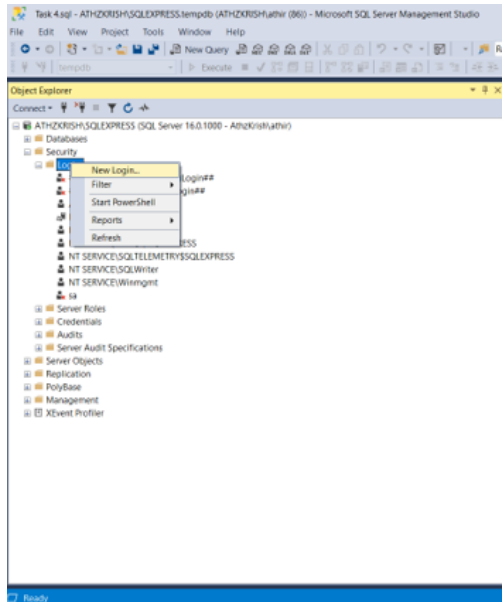


- Select each table in the schema and grant select and update permissions to the Administrator user as below(No delete permission granted to avoid unwanted data loss):

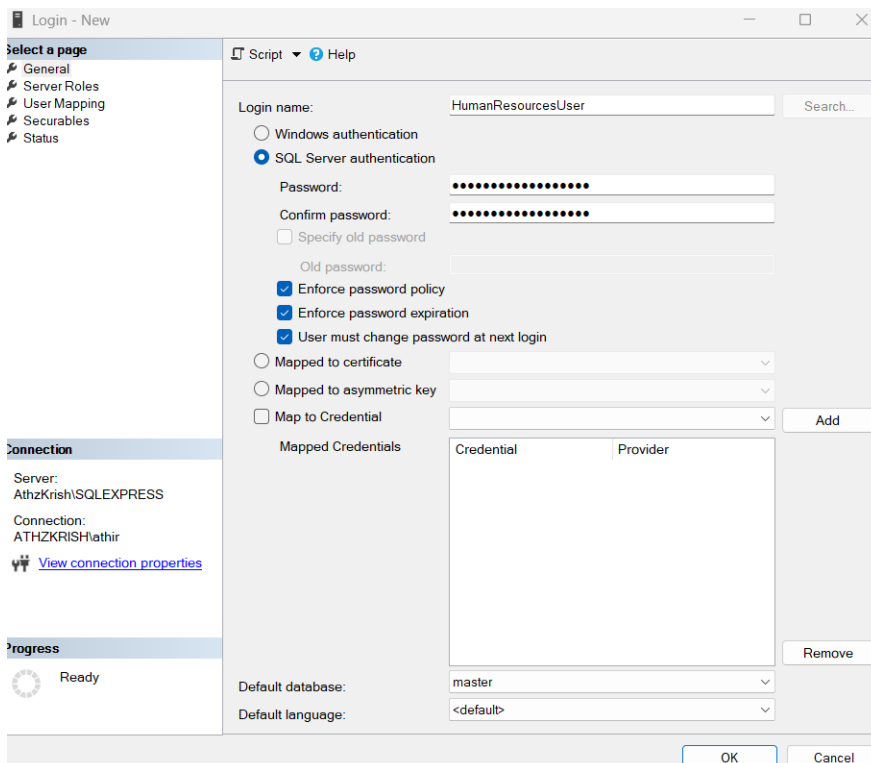


SETTING UP HUMAN RESOURCES USER ACCOUNT BY CREATING SERVER LOGIN AND DATABASE USER:**I. Create Login to the My SQL Server Instance -Server Level Login**

1. Log in to My SQL Server.
2. Go to Security of My SQL Server Instance-> Login and click on New Login option.



New Login creation window opens as below:

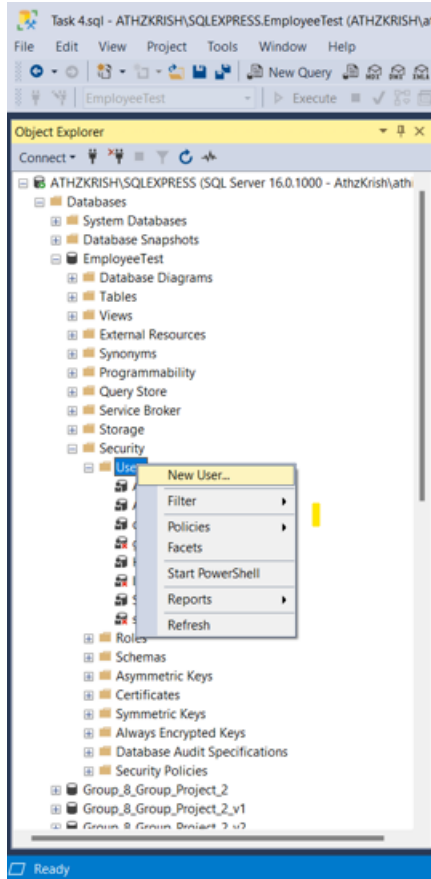


3. Enter the Login Name.
4. Select SQL Server authentication option(For creating logins with password).
5. Enter new password for the login in Password and Confirm Password textboxes.

6. Click OK .

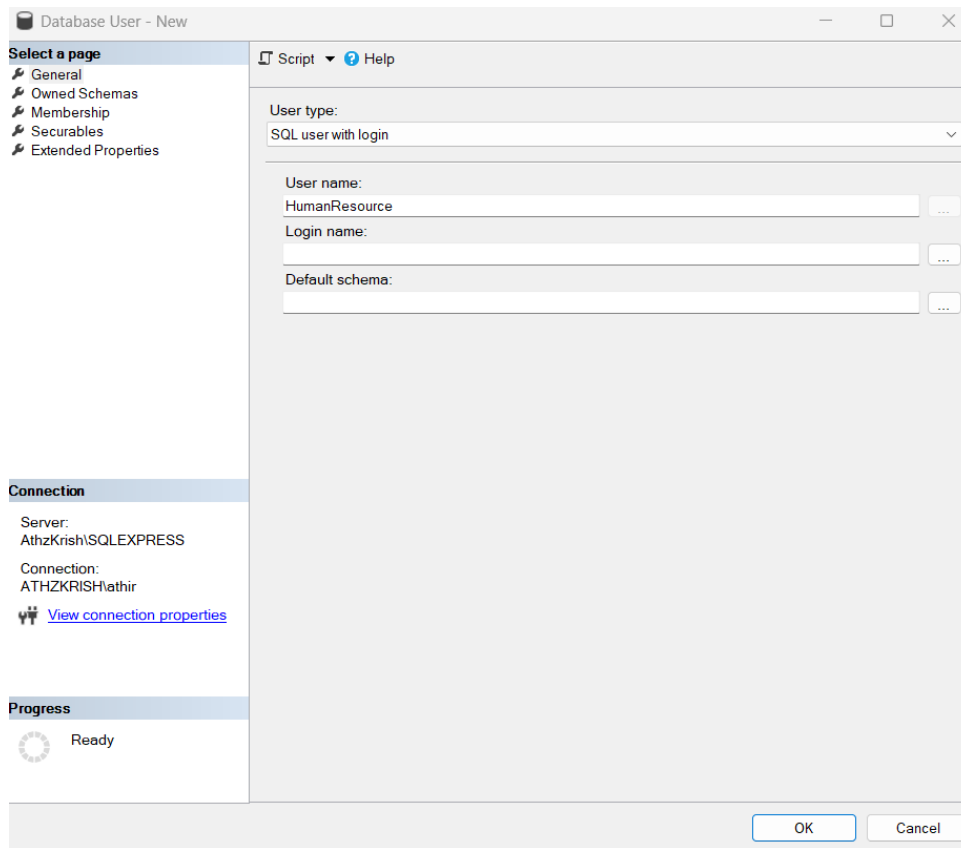
II. Create the Database Level user logins for the Employee Database(Database Name: EmployeeTest)

1. Go to the above database name.
2. Go to Security->User and click on New User option.

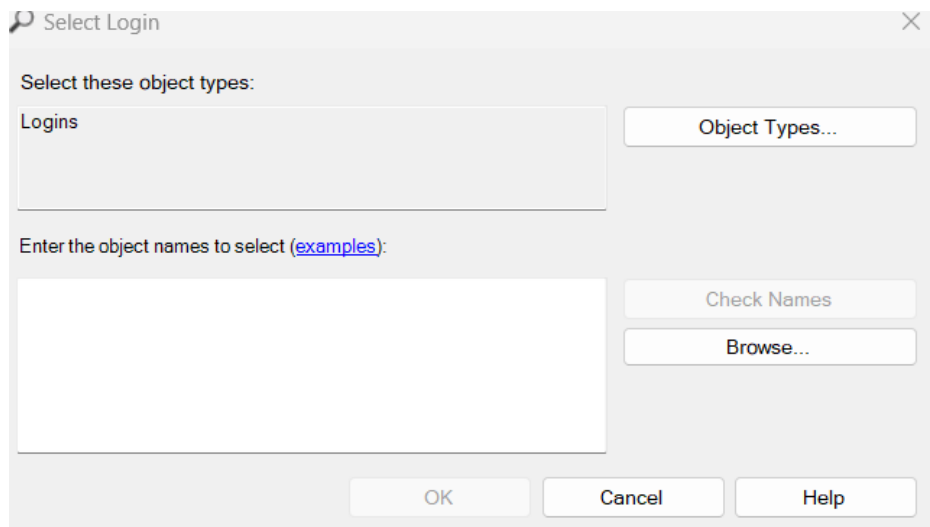


New User creation window opens:

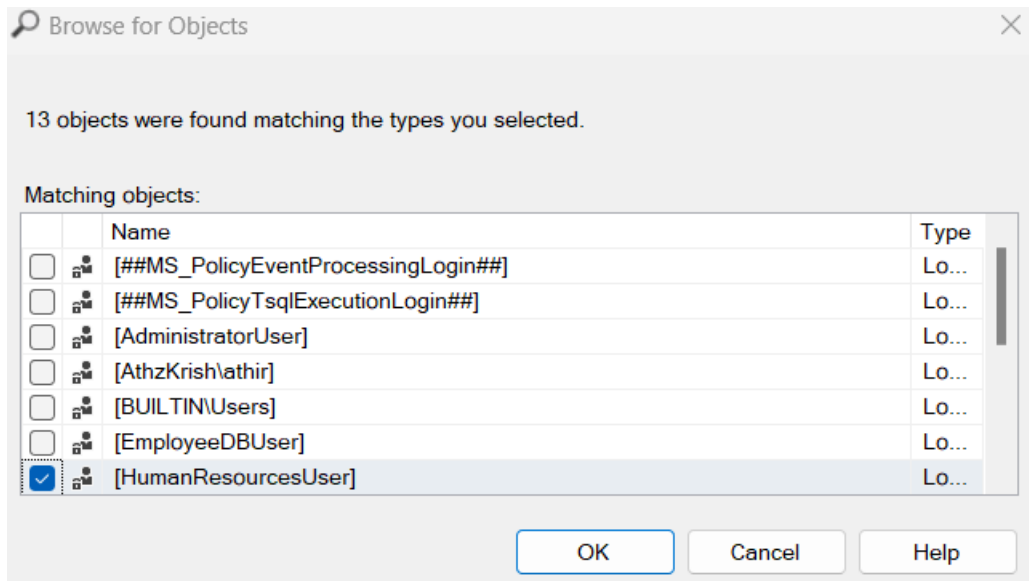
3. Enter the User Name.
Here, I'm creating the Human Resources User Account. Hence, given the user name as 'HumanResource'. Click on the ellipsis(...)next to the Login name textbox to select the server login for this new HumanResource user



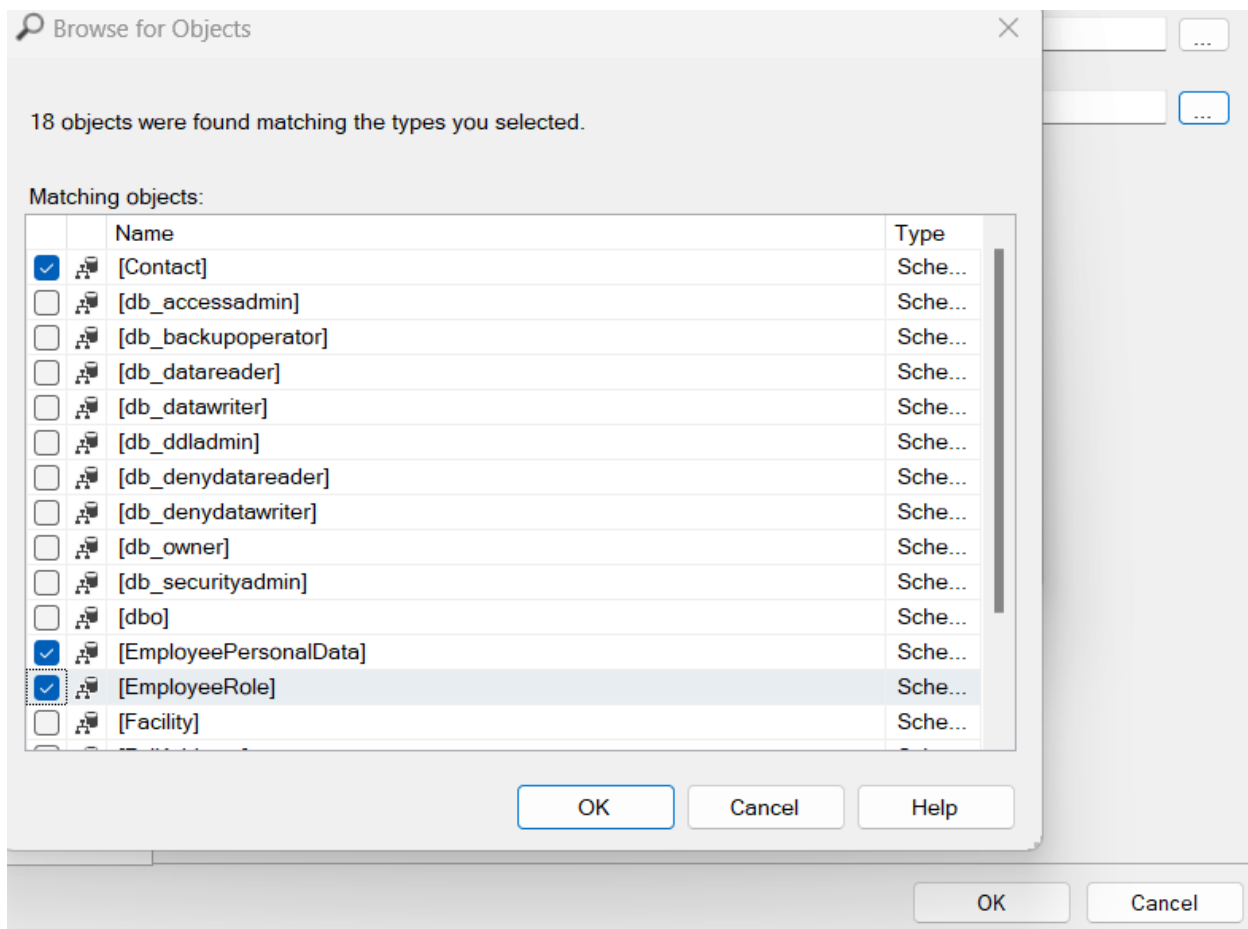
The below opens:



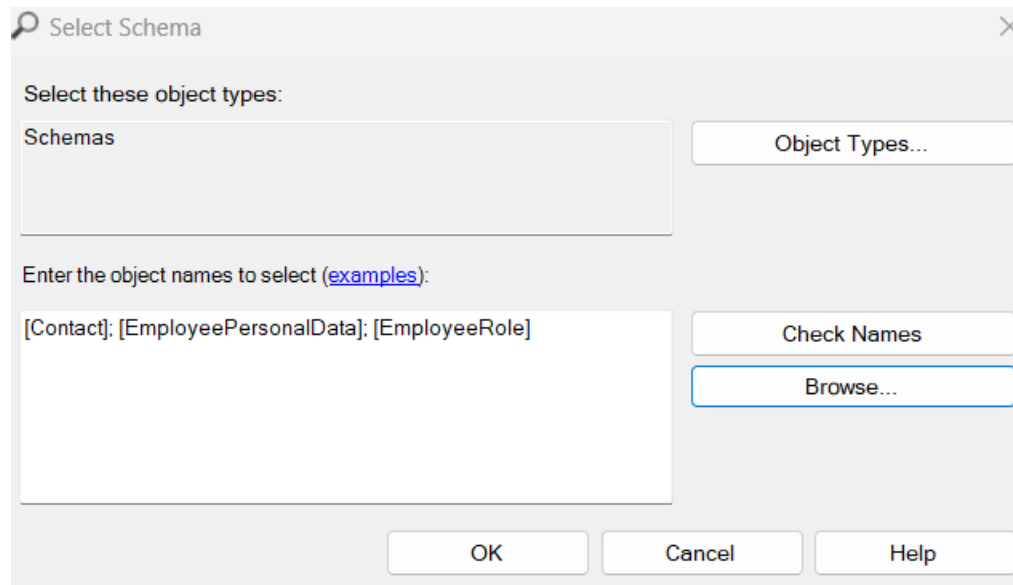
4. Click on the Browse button to select the Server Login for this user. Select the HumanResourcesUser Object from the list and Click OK. (The My SQL server login created for the Human Resource User Account of the Employee Database)



5. Select the schemas: EmployeeRole(Schema 1), EmployeePersonalData(Schema 2) and Contact(Schema 4):



6. Click OK.
7. Selected schemas are added to the object names. Click OK.



Select Schema

Select these object types:

Schemas

Object Types...

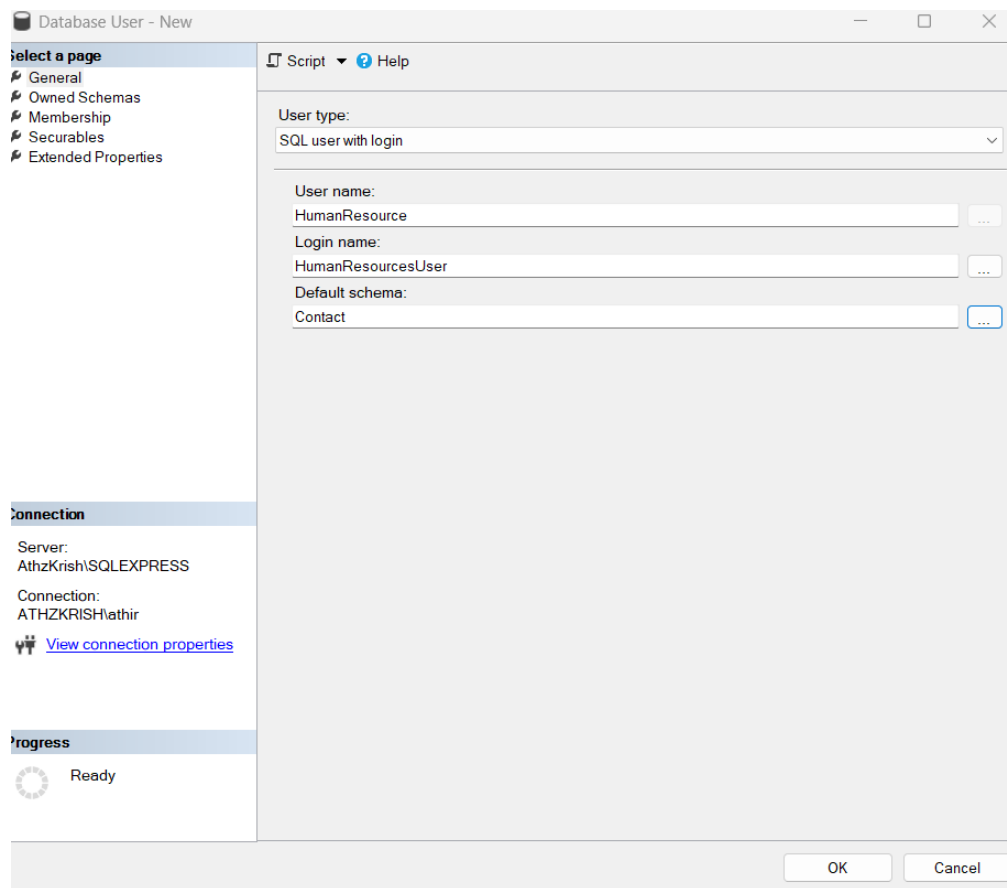
Enter the object names to select ([examples](#)):

[Contact]: [EmployeePersonalData]: [EmployeeRole]

Check Names

Browse...

OK Cancel Help



Database User - New

Select a page

- General
- Owned Schemas
- Membership
- Securables
- Extended Properties

Script Help

User type:

SQL user with login

User name:

HumanResource

Login name:

HumanResourcesUser

Default schema:

Contact

Connection

Server:

AthzKrish\SQLEXPRESS

Connection:

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[View connection properties](#)

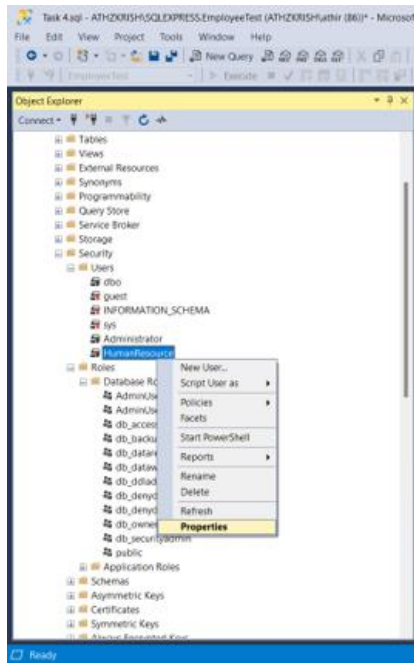
Progress

Ready

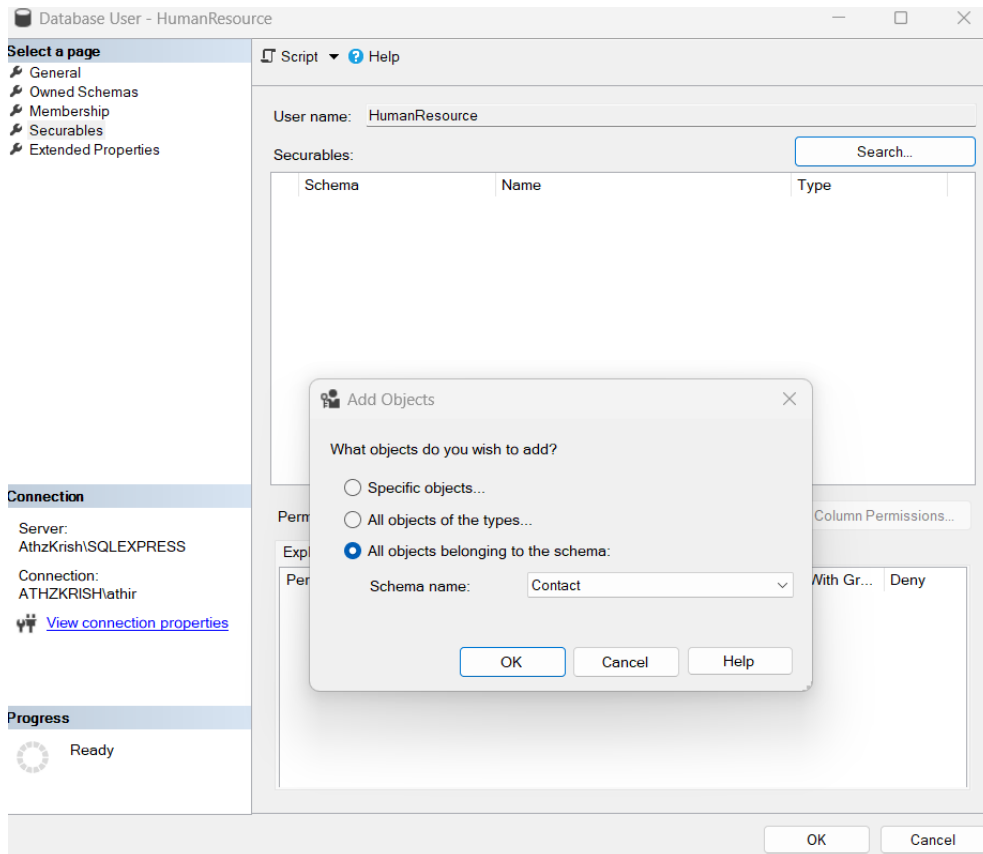
OK Cancel

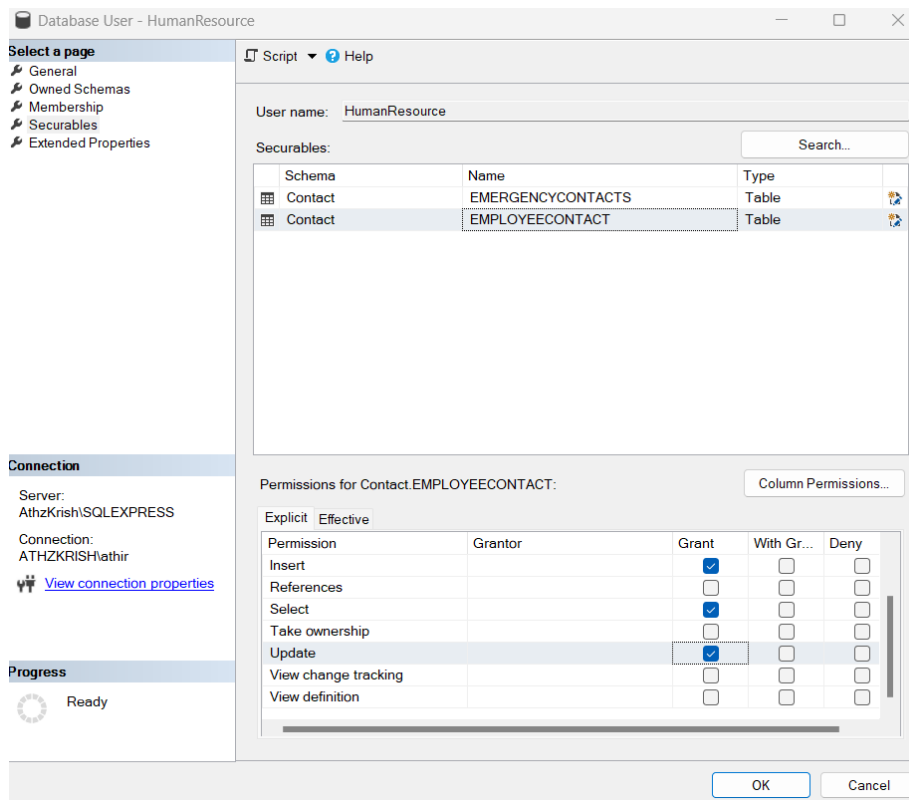
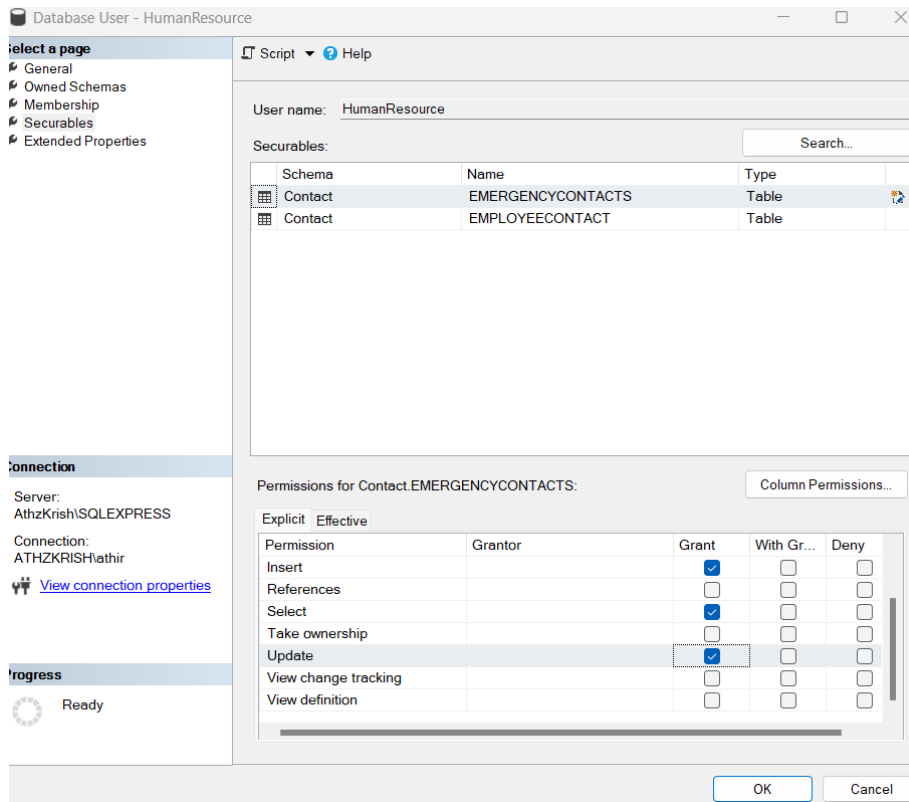
III. Adding Permissions for Human Resources User Account

1. Go back to Database name ->Security->Users->HumanResource and click on Properties.

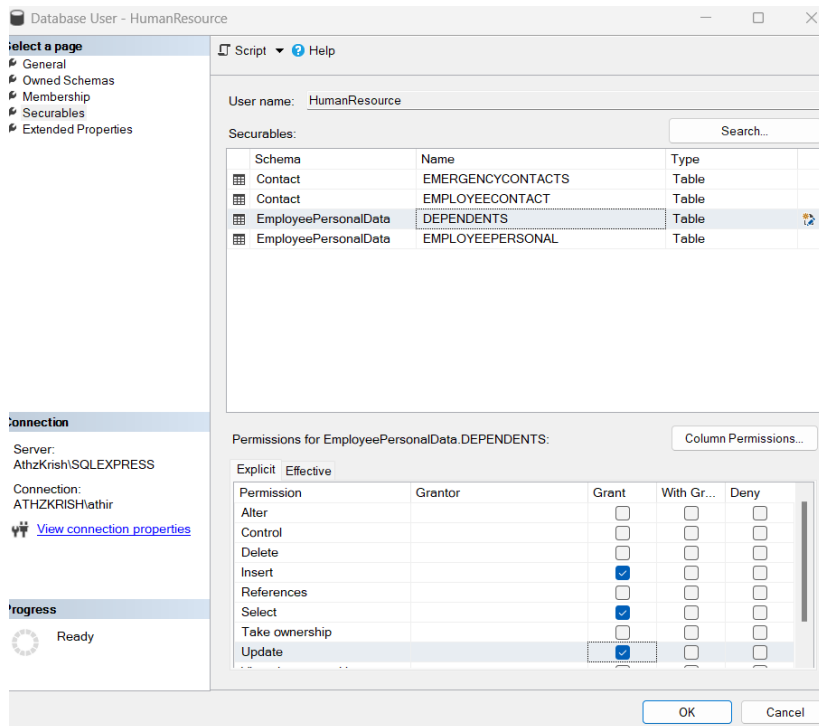
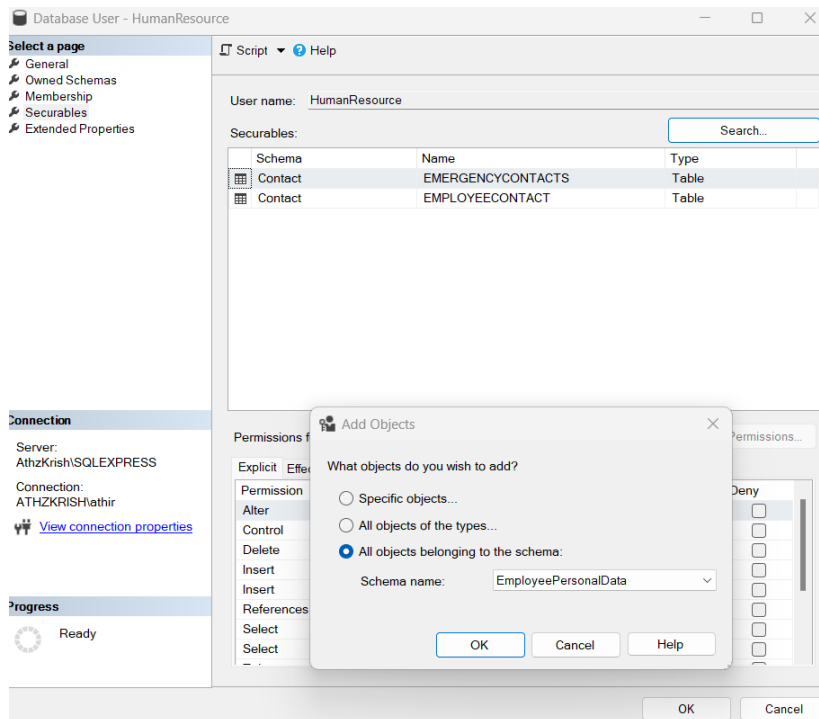


2. Select Securables from the left and click on the Search button:
3. **Select Contact schema and grant insert, update and select permissions to all tables in Contact schema**

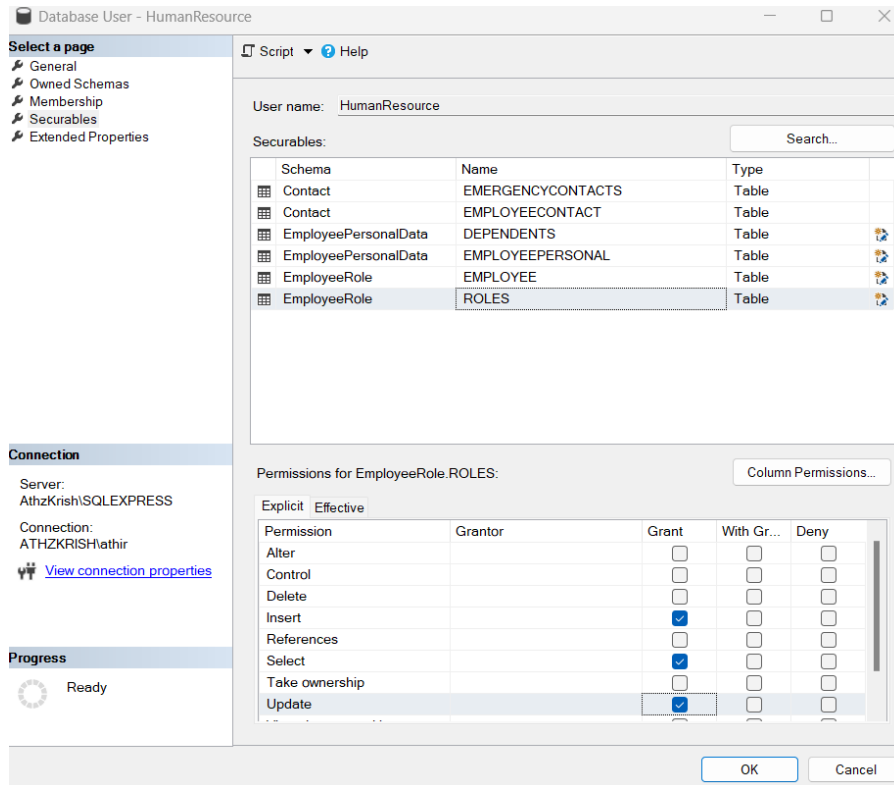
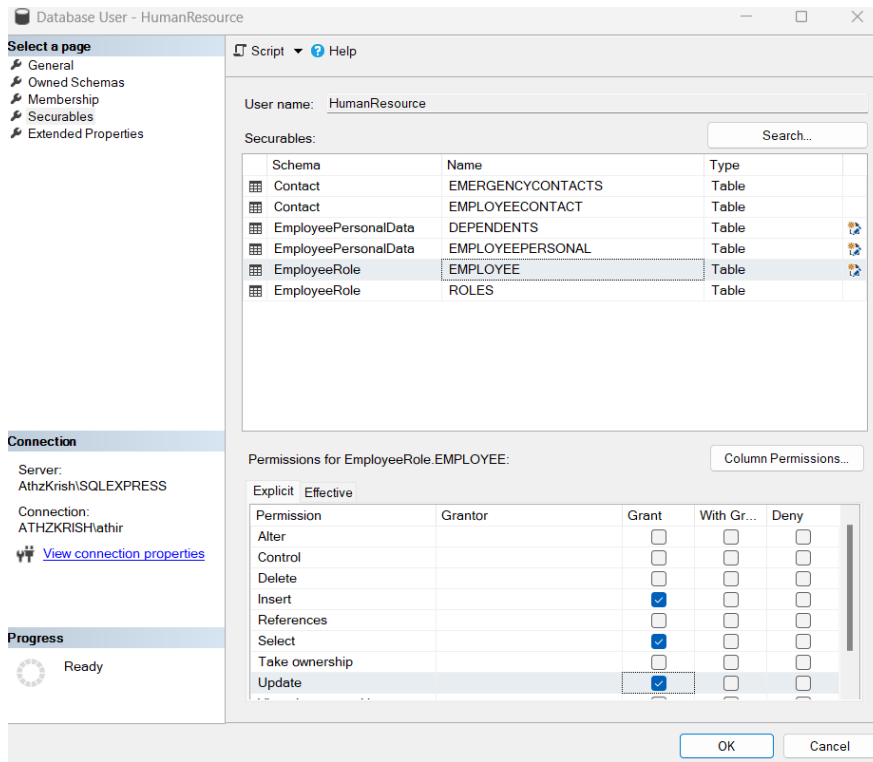




4. Similarly grant insert, update, and select permissions to all tables in EmployeePersonalData schema.



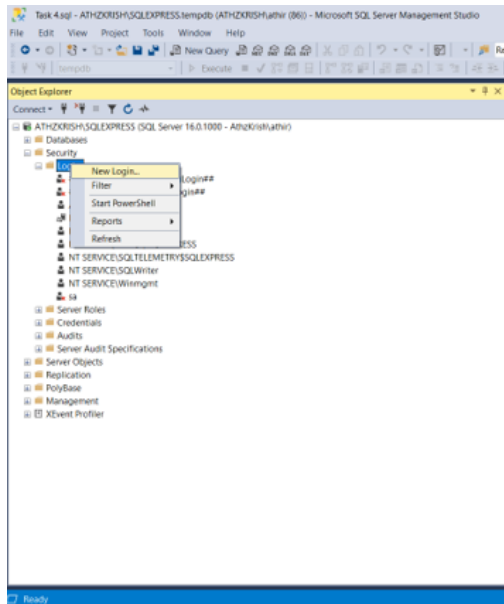
5. *Similarly grant insert, update, and select permissions to all tables in the EmployeeRole schema.*



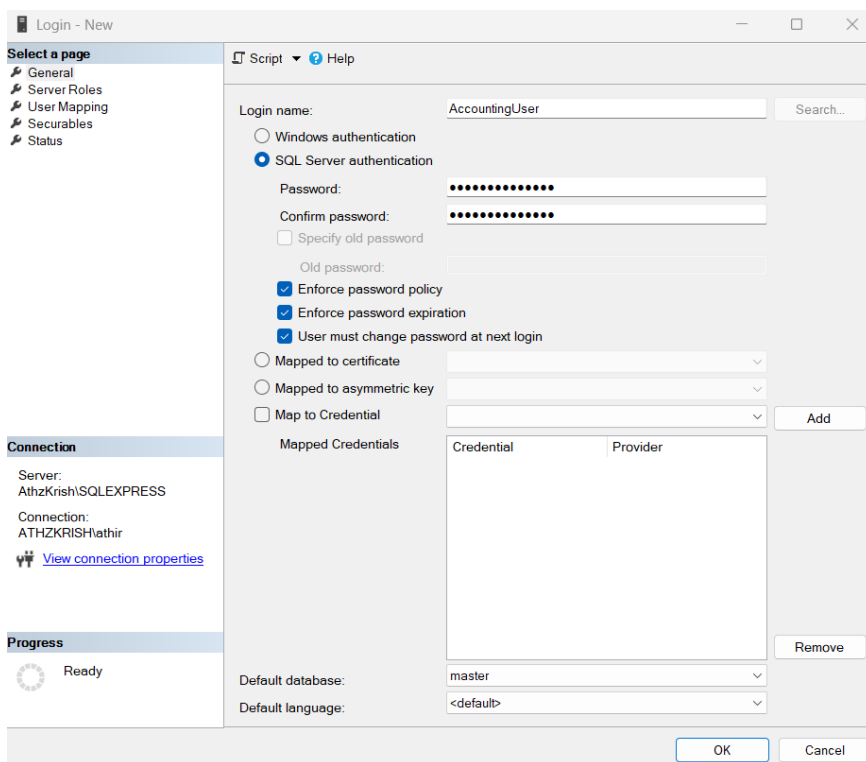
SETTING UP ACCOUNTING USER ACCOUNT BY CREATING A SERVER LOGIN AND DATABASE USER:

I. Create Login to the My SQL Server Instance -Server Level Login

1. Log in to My SQL Server.
2. Go to Security of My SQL Server Instance-> Login and click on New Login option.



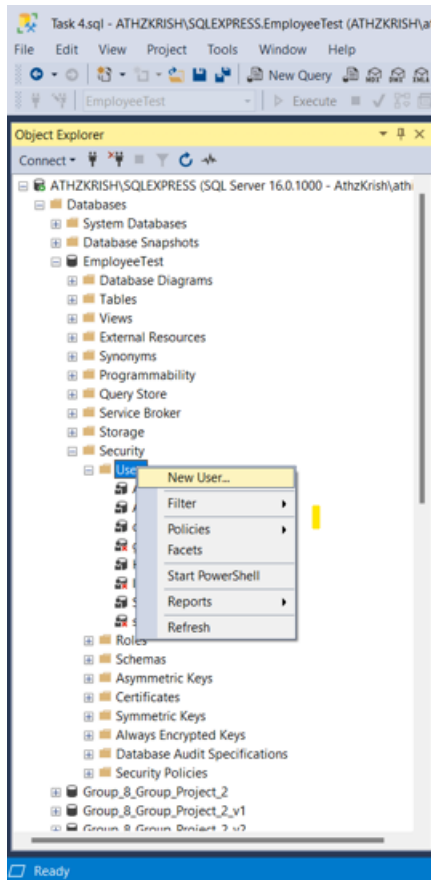
New Login creation window opens as below:



3. Enter the Login Name.
4. Select SQL Server authentication option(For creating logins with password).
5. Enter new password for the login in Password and Confirm Password textboxes.
6. Click OK .

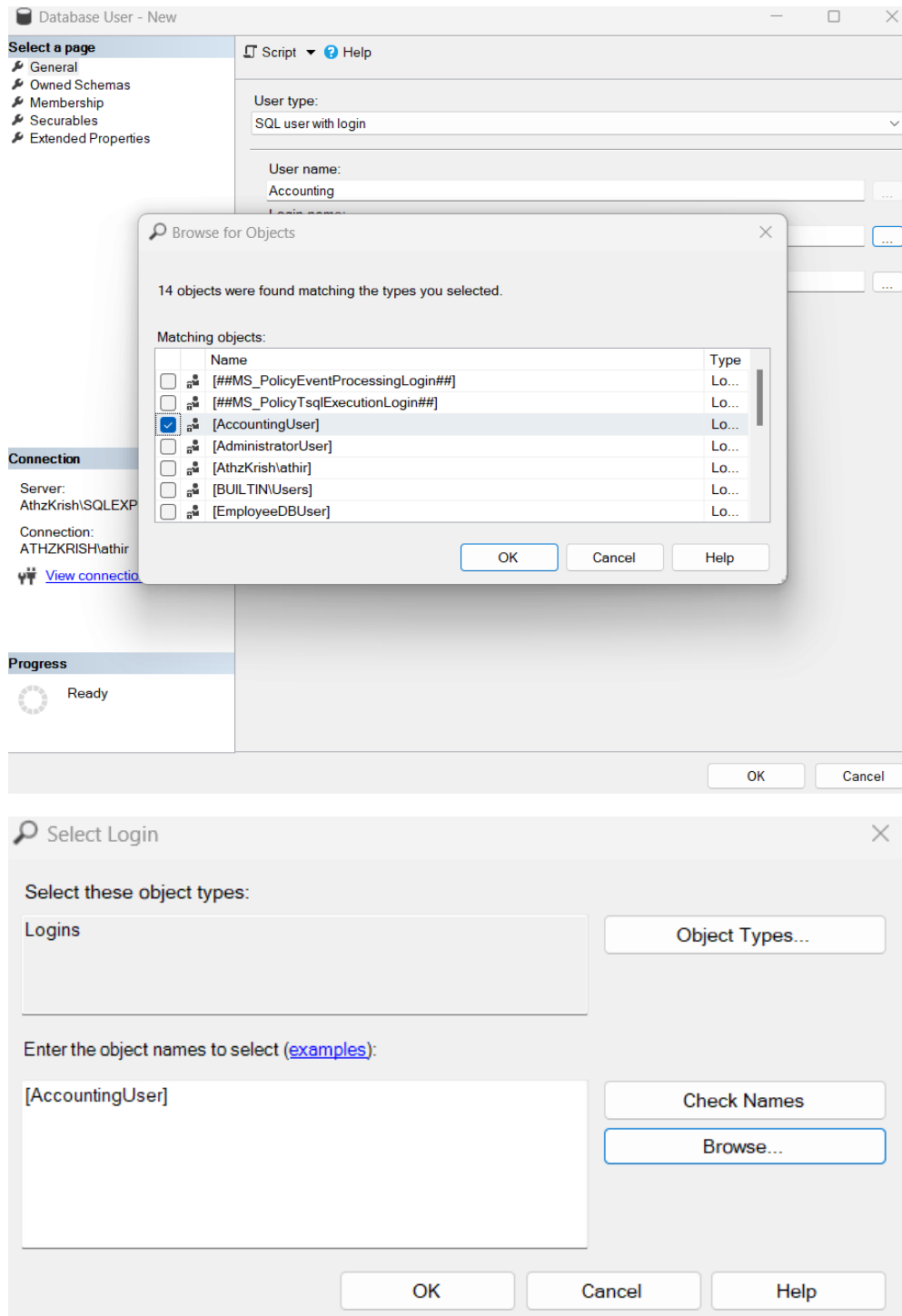
II. Create the Database Level user logins for the Employee Database(Database Name: EmployeeTest)

1. Go to the above database name.
2. Go to Security->User and click on New User option.

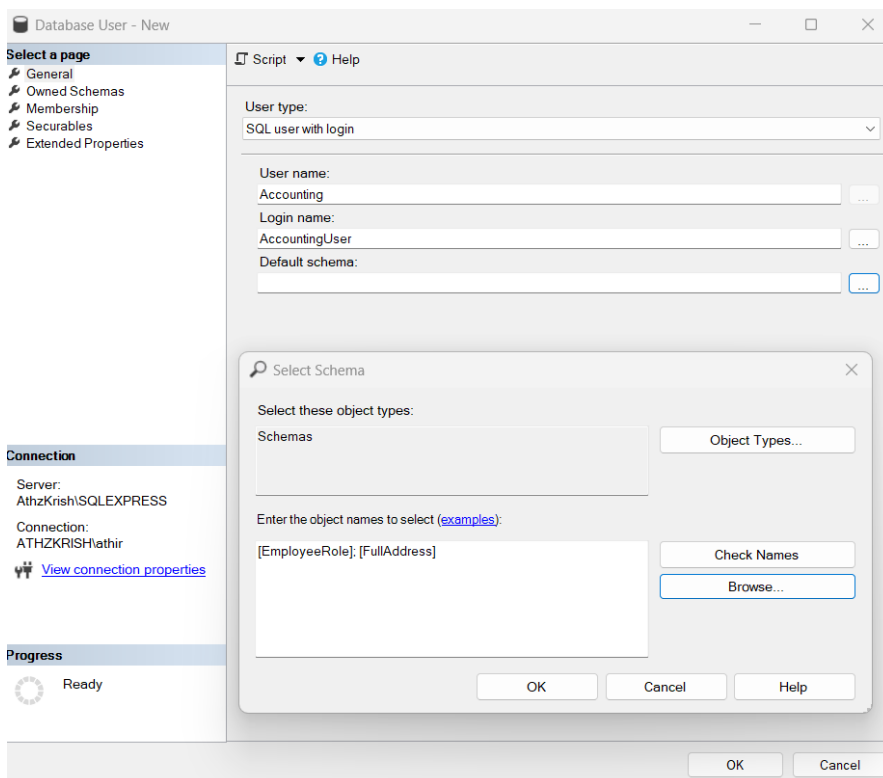
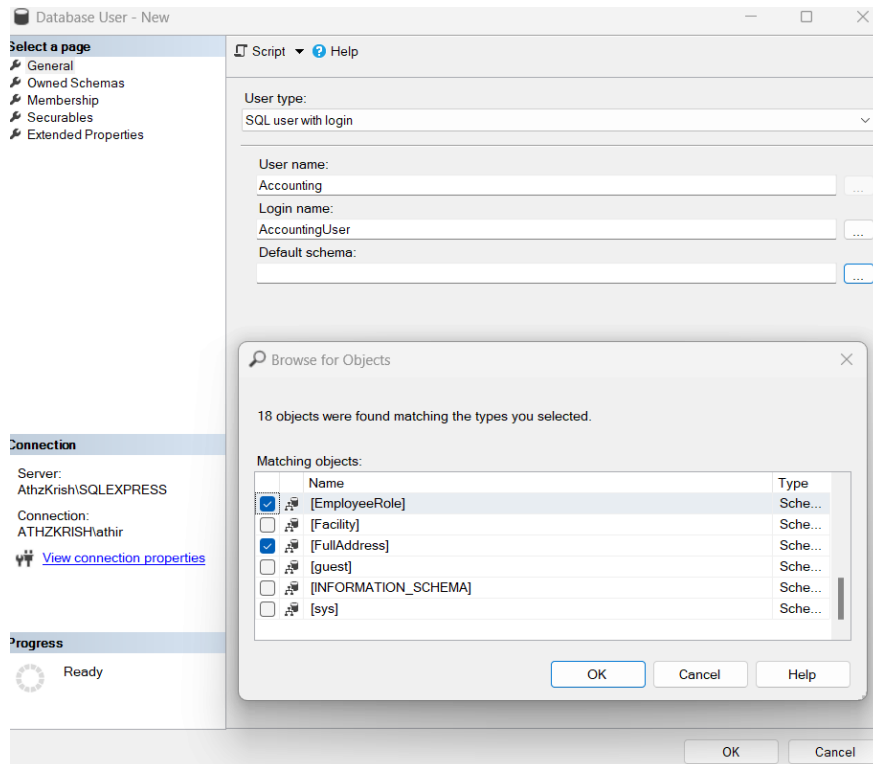


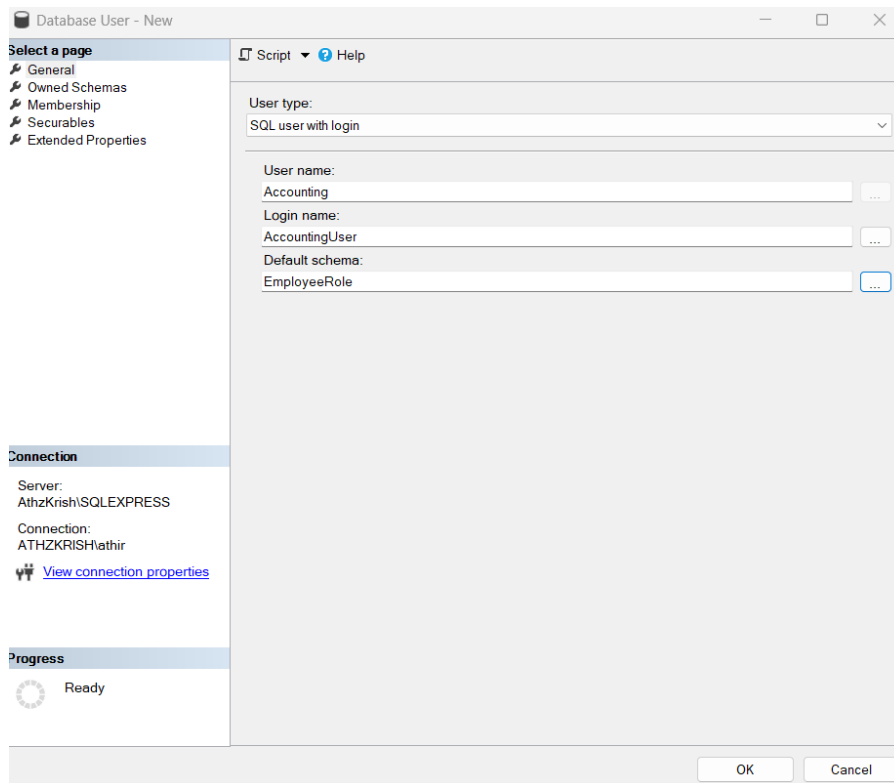
New User creation window opens:

3. Enter the User Name.
Here, I'm creating the Human Resources User Account. Hence, given the user name as 'Accounting.'
4. Click on the ellipsis(...)next to the Login name textbox to select the server login for this new Accounting user



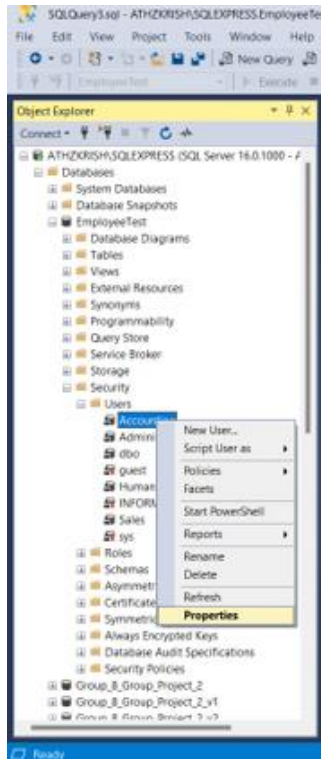
5. Select the schemas EmployeeRole and FullAddress and click OK.



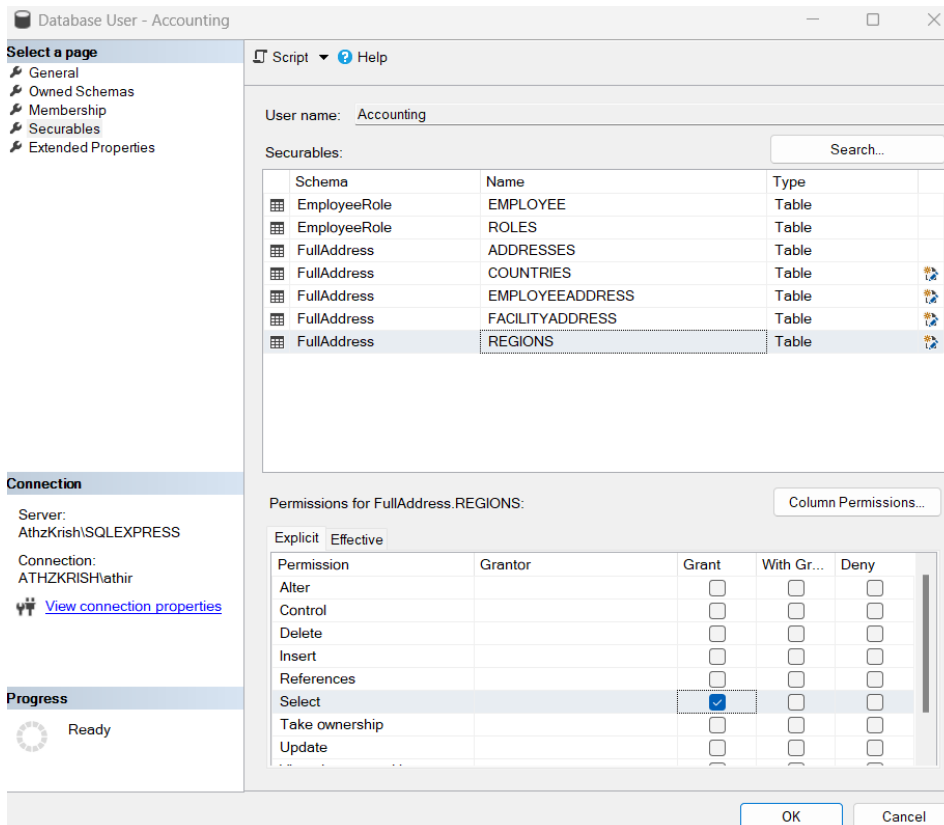
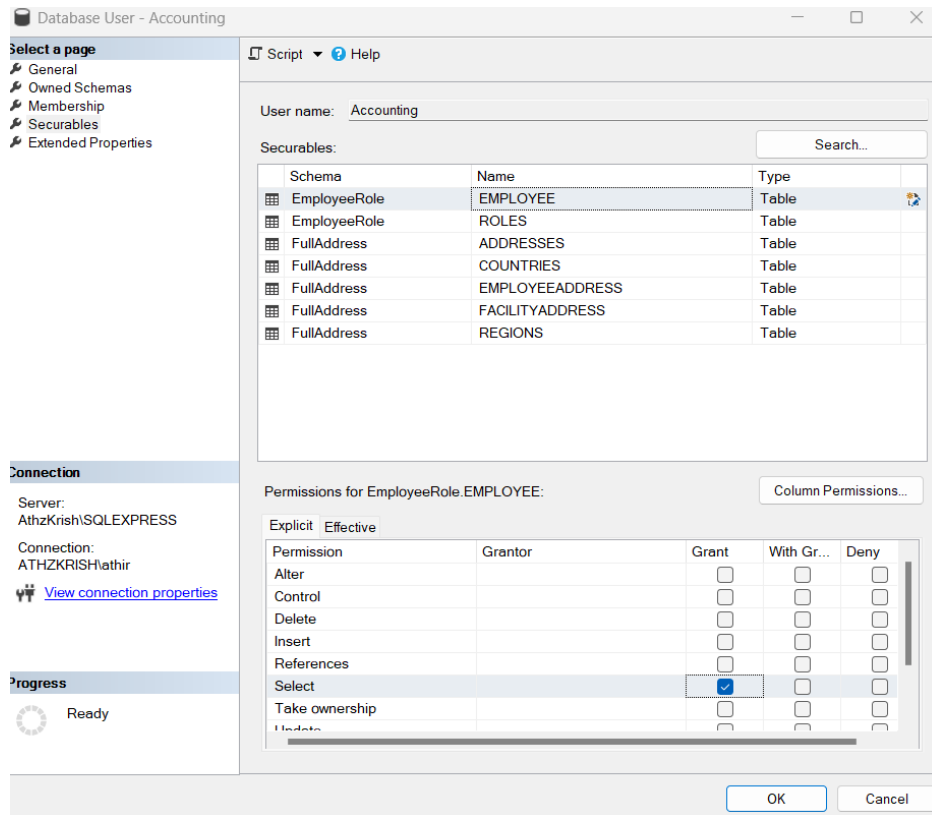


III. Adding Permissions for Accounting User Account

1. Go back to Database name ->Security->Users->Accounting and click on Properties.



2. Grant only select permissions to all tables in EmployeeRole and Address schema.

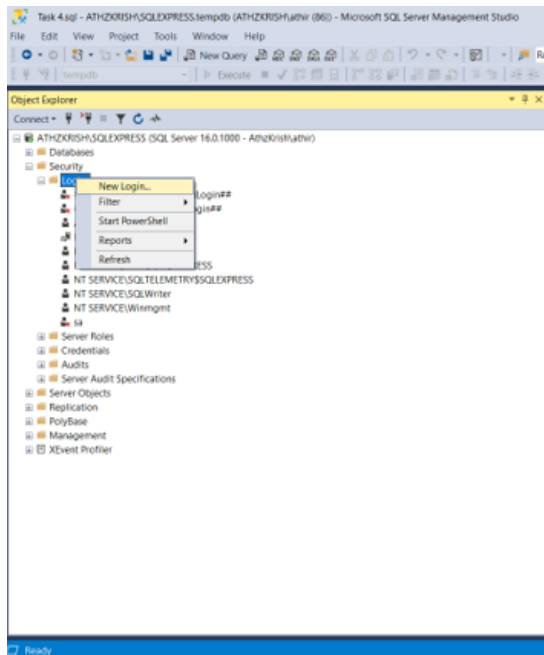


SETTING UP SALES USER ACCOUNT BY CREATING SERVER LOGIN AND DATABASE USER:

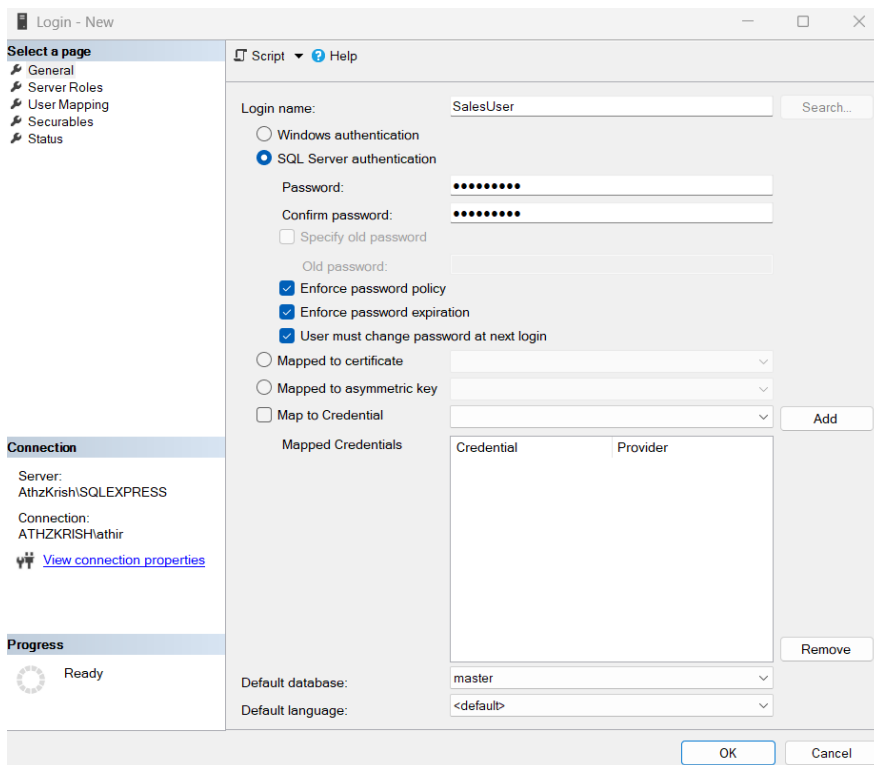
I. Create Login to the My SQL Server Instance -Server Level Login

1. Log in to My SQL Server.

2. Go to Security of My SQL Server Instance-> Login and click on New Login option.



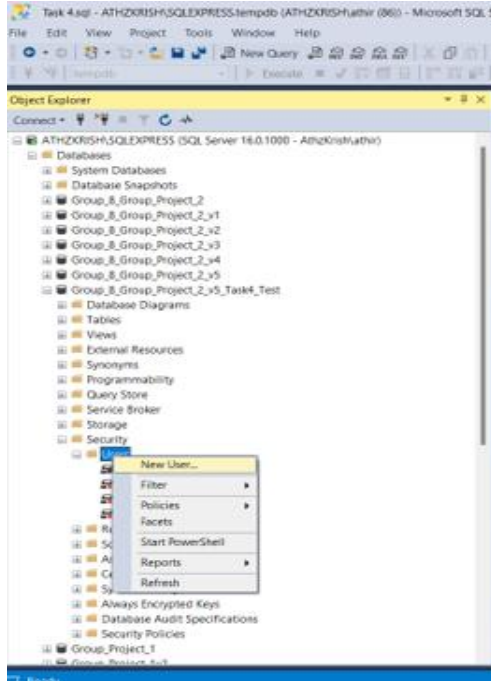
New Login creation window opens as below:



7. Enter the Login Name.
8. Select SQL Server authentication option(For creating logins with password).
9. Enter new password for the login in Password and Confirm Password textboxes.
10. Click OK .

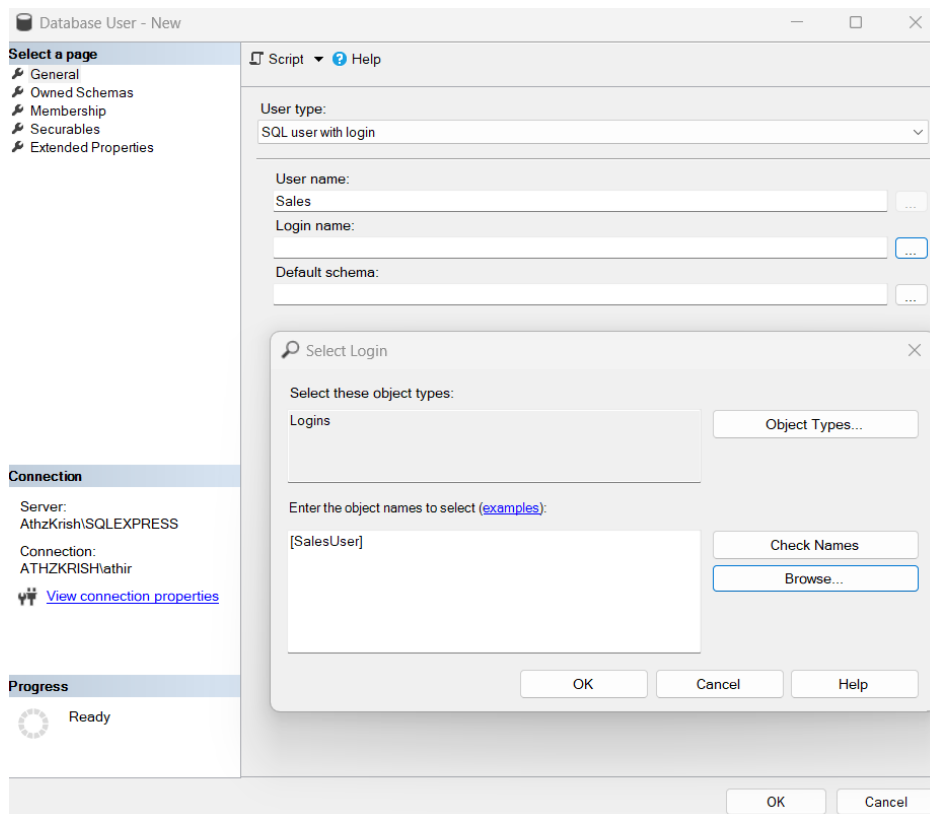
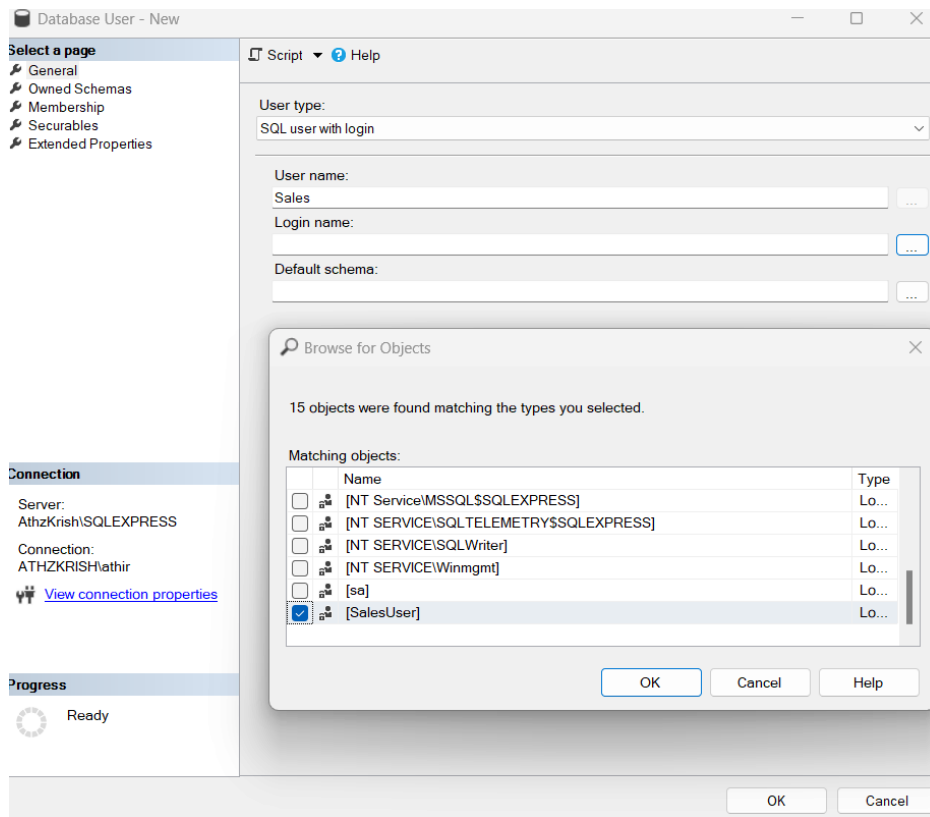
II. Create the Database Level user logins for the Employee Database(Database Name: EmployeeTest)

1. Go to the above database name.
2. Go to Security->User and click on New User option.

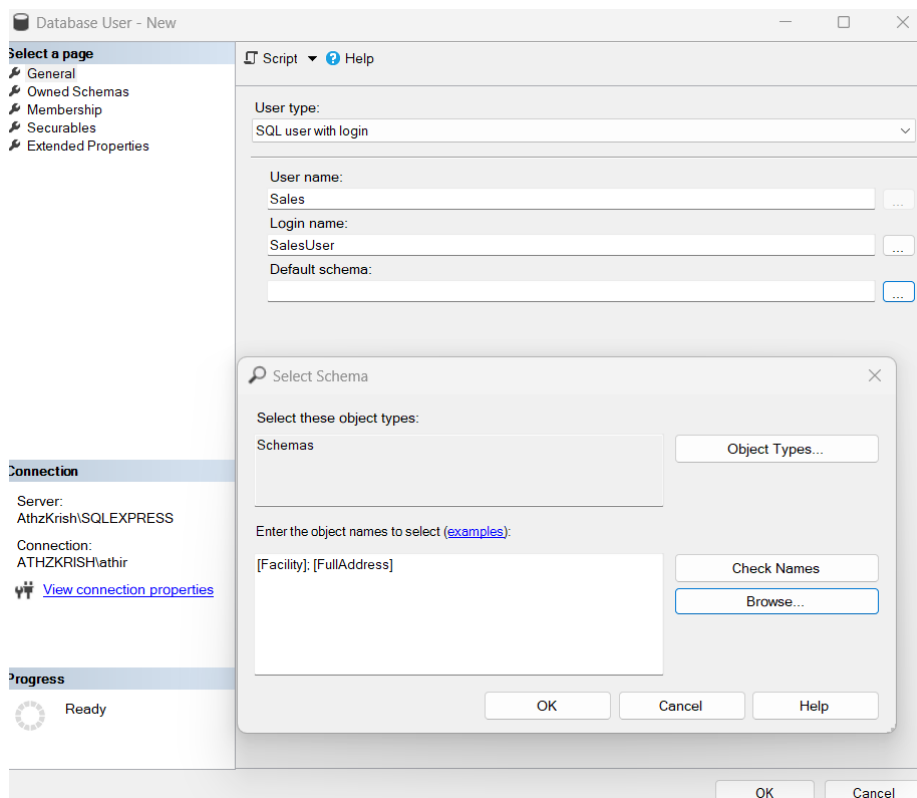
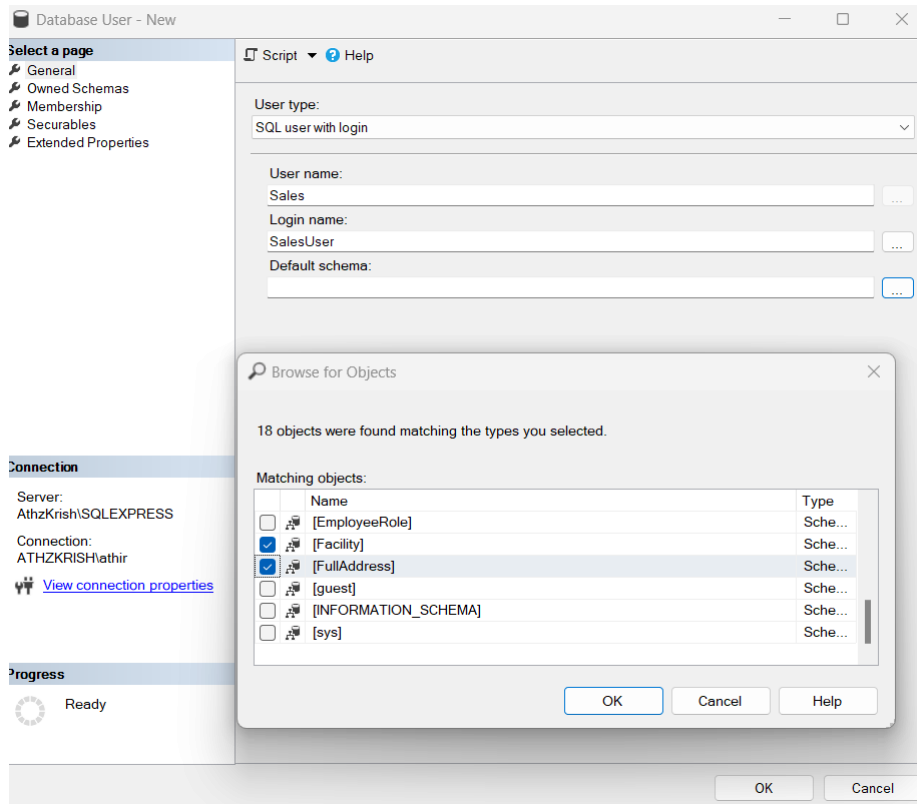


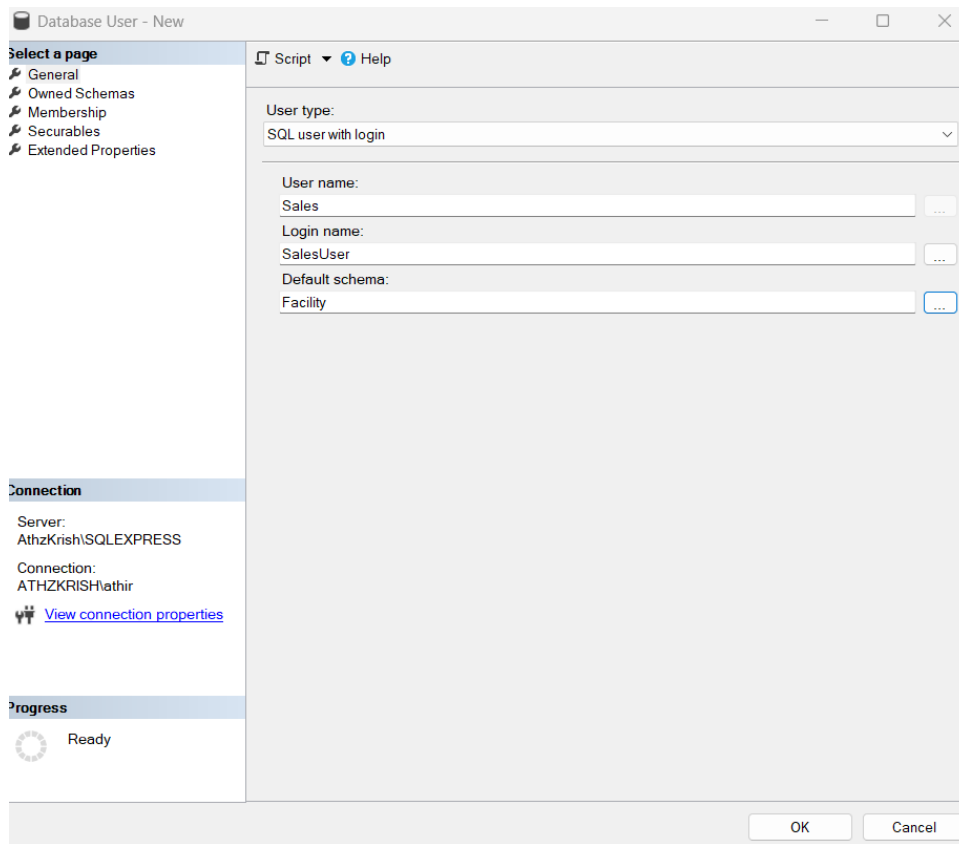
New User creation window opens:

3. Enter the User Name.
Here, I'm creating the Sales User Account. Hence, given the user name as 'Sales'.
Click on the ellipsis(...)next to the Login name textbox to select the server login for this new Sales user



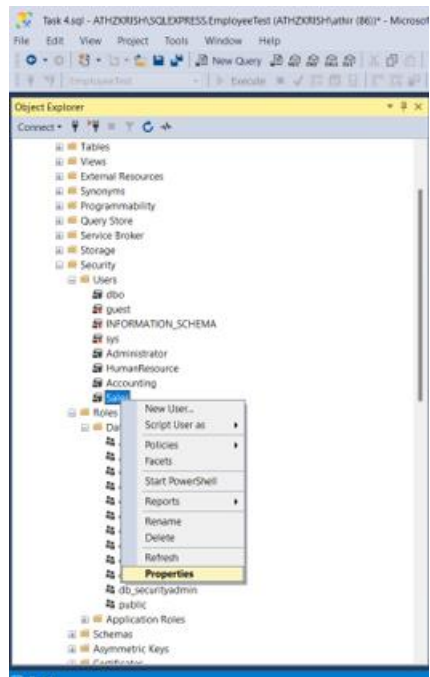
4. Select Facility and Address Schema.



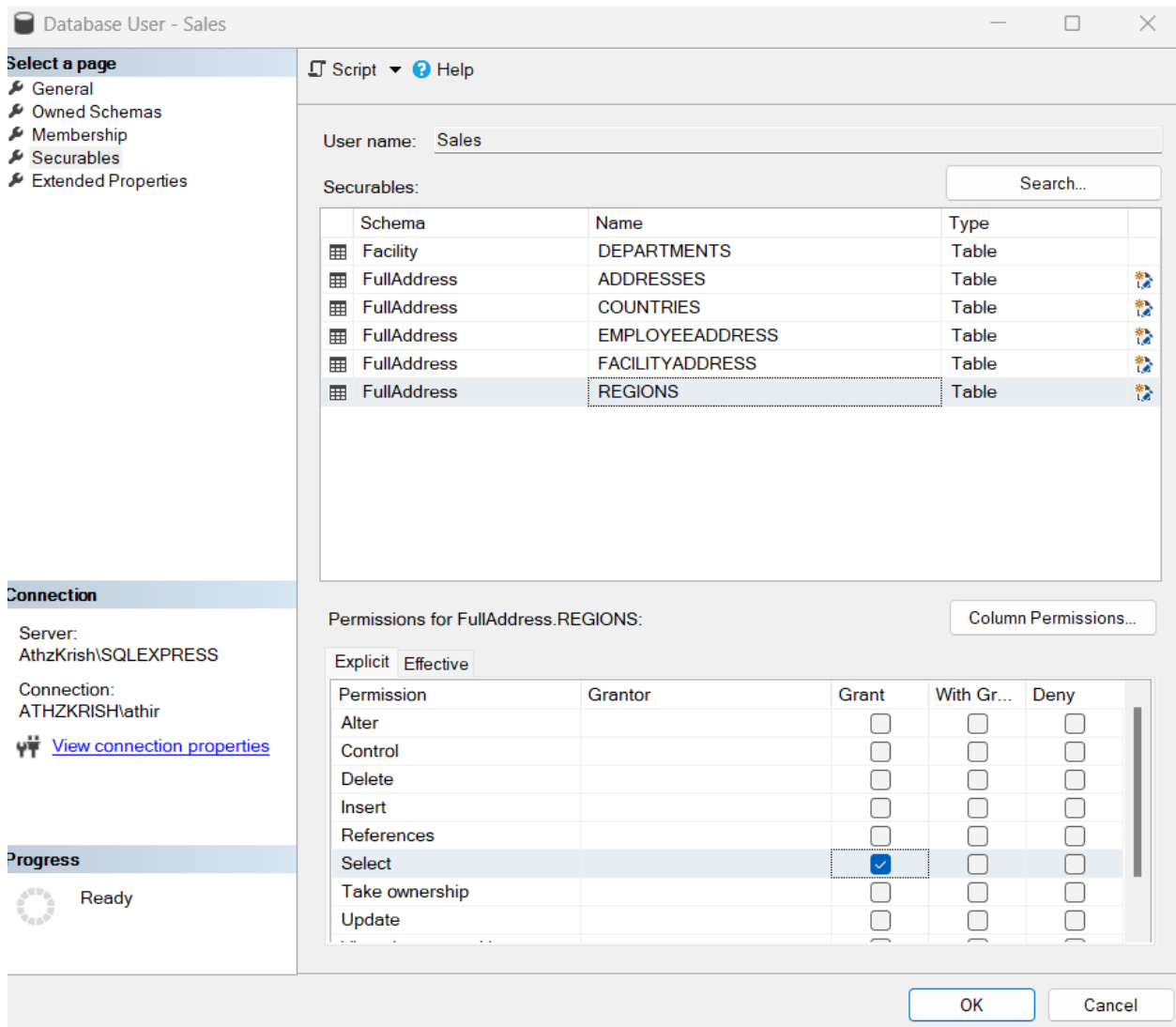


III. Adding Permissions for Administrator User Account

1. Go back to Database name ->Security->Users->Administrator and click on Properties.



2. ***Grant only select permissions to all tables in Facility and Address schema.***

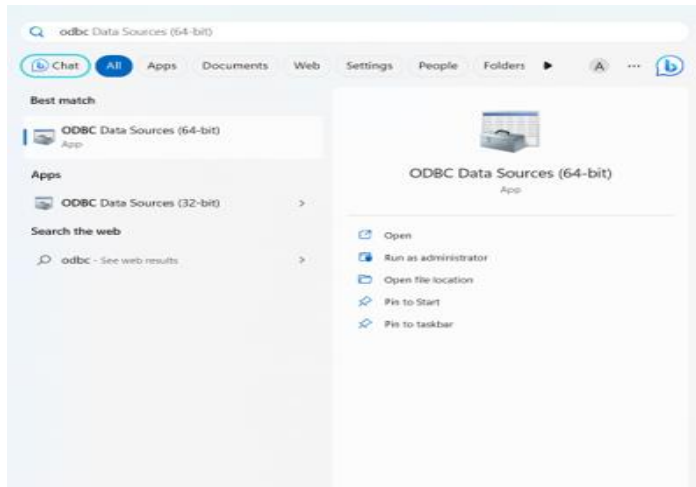


```
--Assuming the probation period is 6 months for all employees
CREATE VIEW ProbationEmployees AS
SELECT TRIM(E.FName) + ' '+TRIM(E.LName) AS 'Employees in Probation Period'
FROM EmployeeRole.EMPLOYEE E JOIN EmployeeRole.ROLES R ON E.RoleID =R.RoleID and
R.ProbationLength <= 6);
```

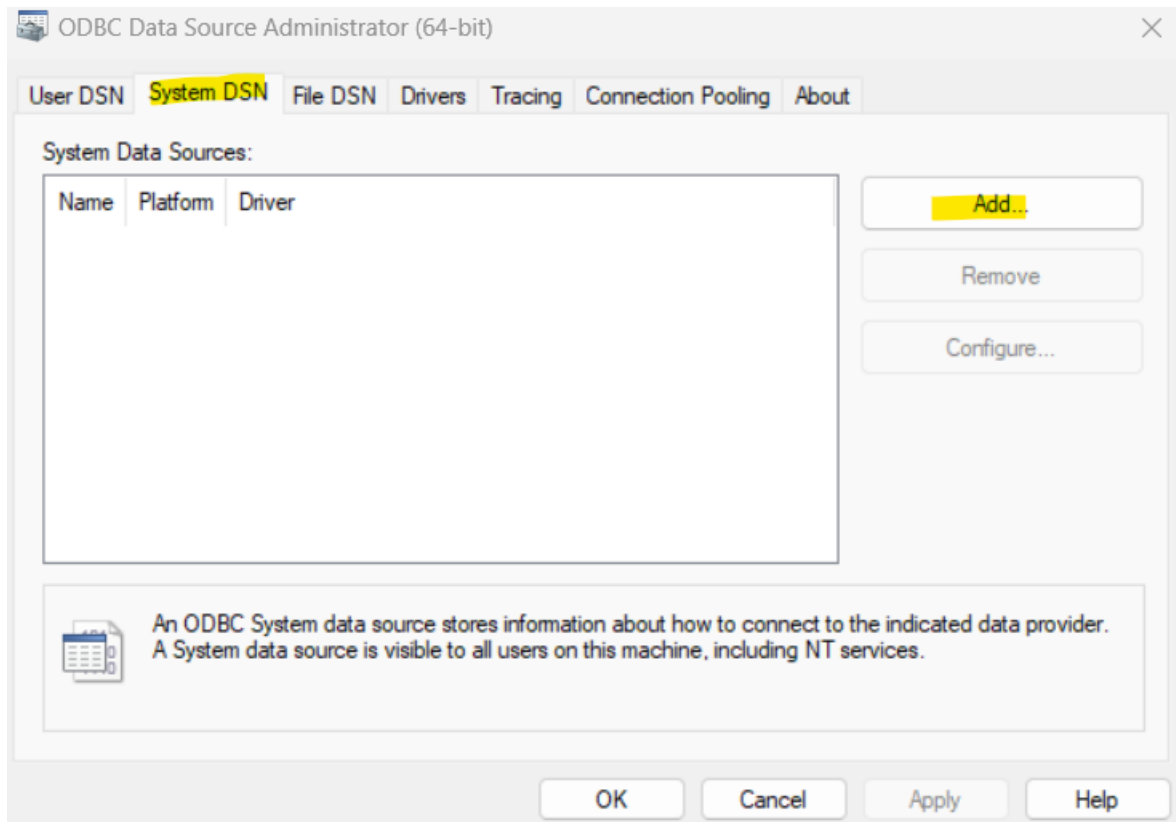
```
SELECT * FROM ProbationEmployees;
```

EMAIL 3:

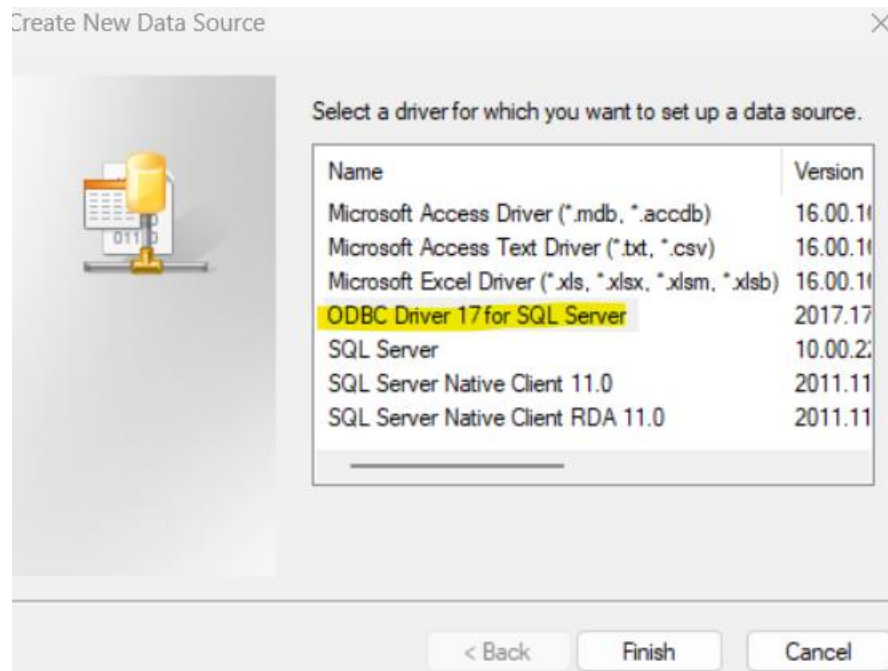
1. Search ODBC Data Sources(64-bit) in Windows Search on your computer. (Note: Use 32-bit if your Windows is 32-bit)



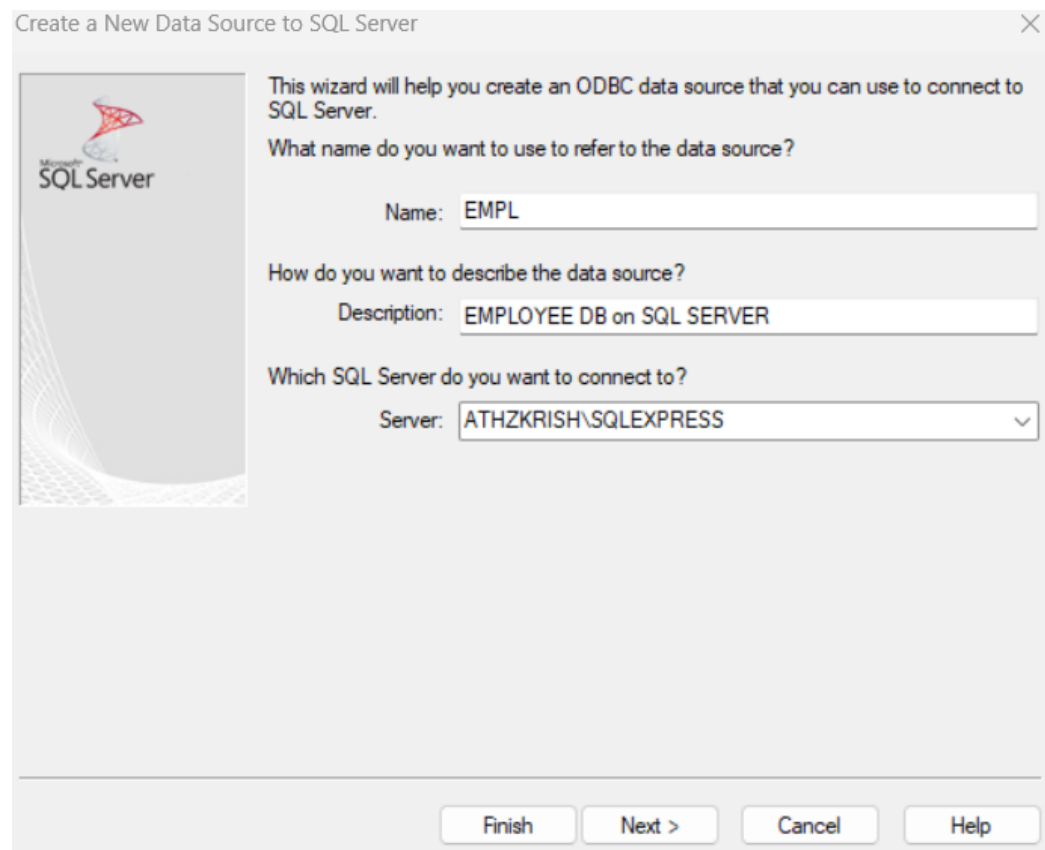
2. Open ODBC Data Source Administrator(64-bit) and go to System DSN Tab, Click Add button.



3. Select ODBC Driver 17 for SQL Server and Click Finish.



4. Create a New Data Source to SQL Server window opens.
5. Enter the Data Source Name, Description, and SQL Server Name you want to connect to as shown below.
6. Click Next.



7. Select With Integrated Windows authentication.

Create a New Data Source to SQL Server



How should SQL Server verify the authenticity of the login ID?

☒ With Integrated Windows authentication.

SPN (Optional):

☐ With Azure Active Directory Integrated authentication.

☐ With SQL Server authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.

☐ With Azure Managed Service Identity authentication.

☐ With Azure Service Principal authentication.

Login ID:

Password:

< Back Next > Cancel Help

8. Change the default database to EmployeeTest DB(Our Employee Database) and click Next.

Create a New Data Source to SQL Server



☒ Change the default database to:

EmployeeTest

Mirror server:

SPN for mirror server (Optional):

☐ Attach database filename:

☒ Use ANSI quoted identifiers.

☒ Use ANSI nulls, paddings and warnings.

Application intent:

READWRITE

☐ Multi-subnet failover.

☒ Transparent Network IP Resolution.

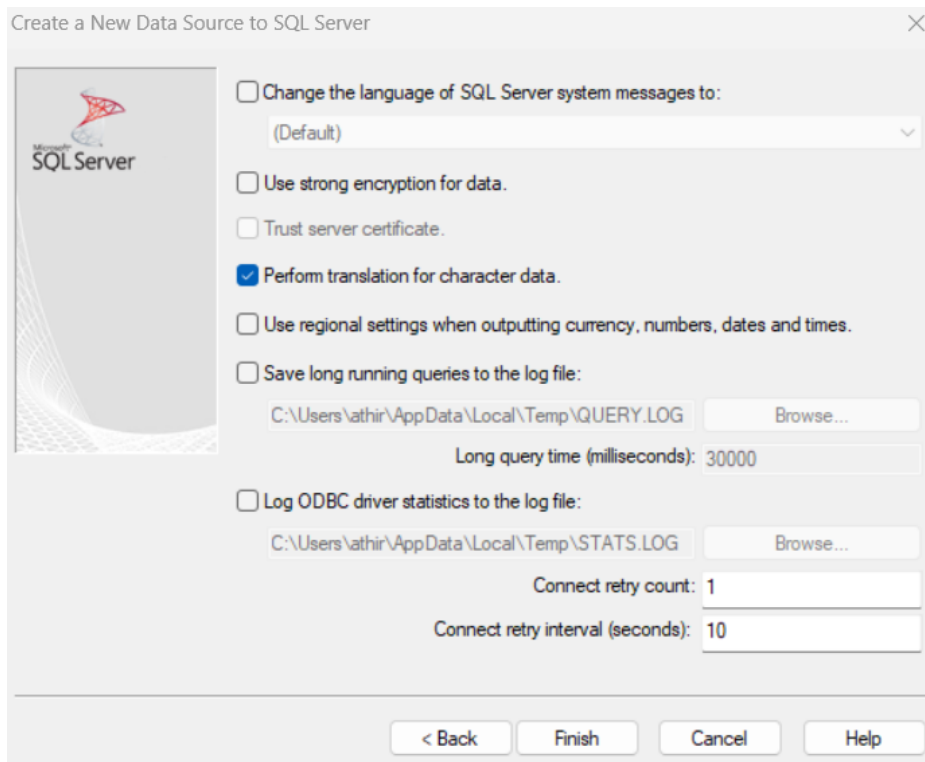
☐ Column Encryption.

Enclave Attestation Info:

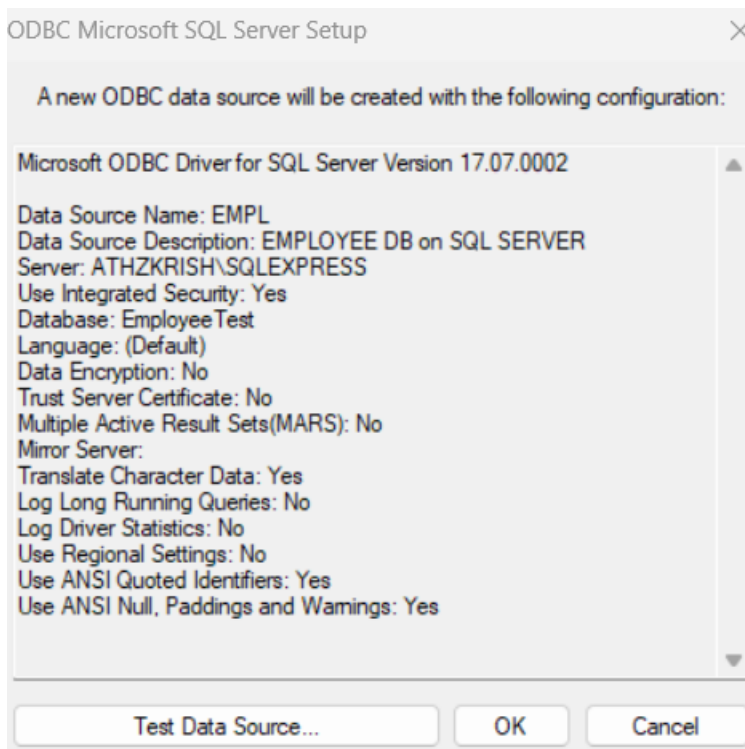
☐ Use FMTONLY metadata discovery.

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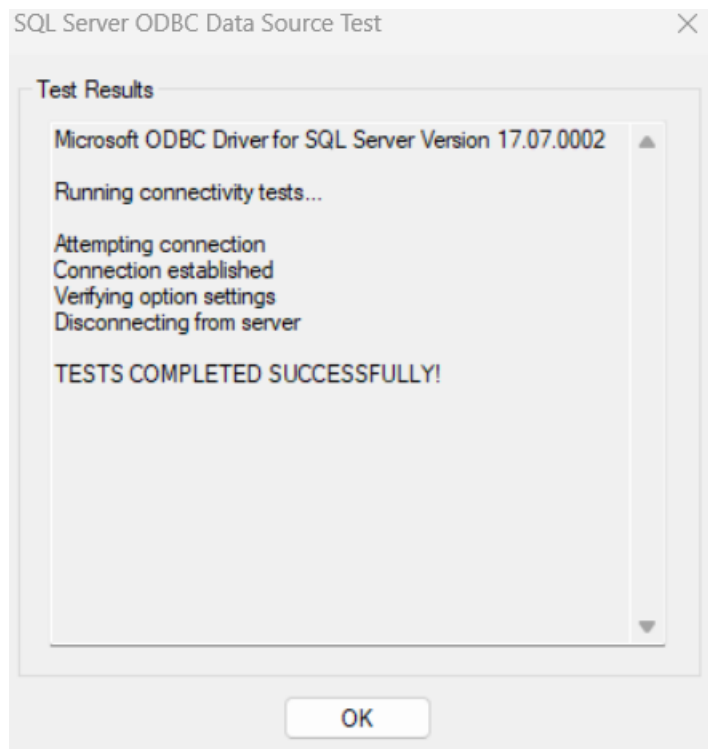
9. Click Finish.



10. Click Test Data Source.



11. ODBC Connection to SQL Server is Successful.



12. New Data Source added in the System DSN.

